Compute Forecast

Romeo Dean | April 2025

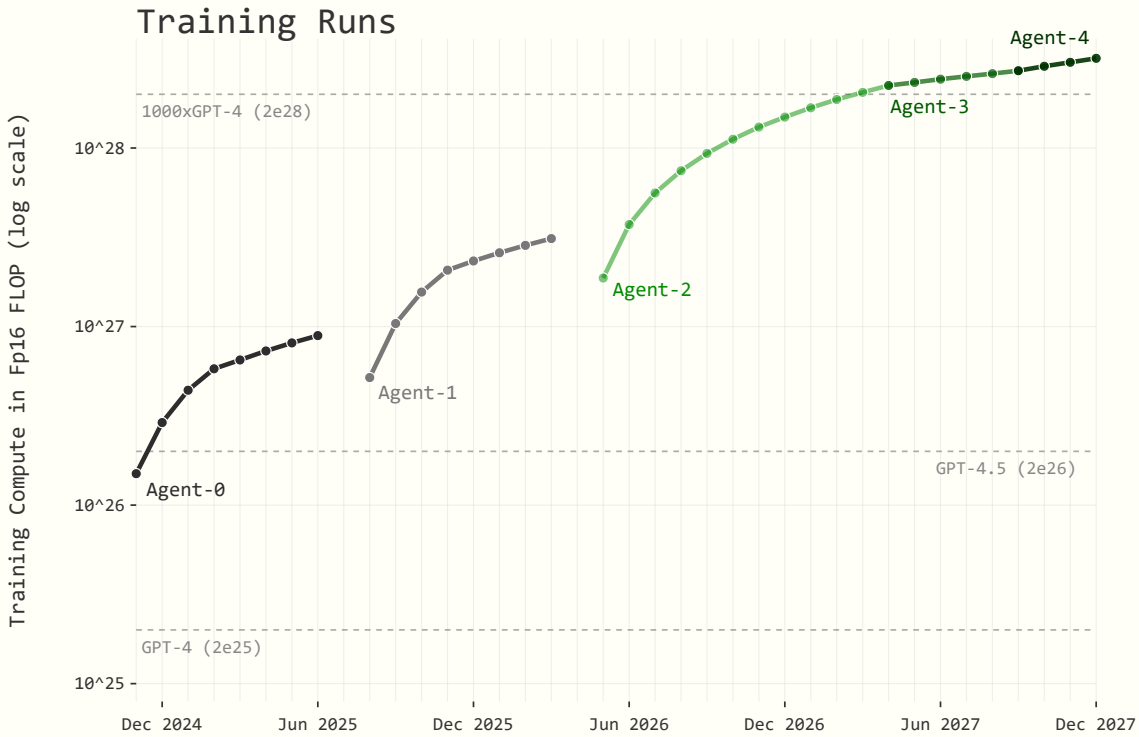


Figure 1: The training runs in AI 2027, informed and justified by our compute production, compute distribution and compute usage sections.

Summary

Compute is one of the key inputs to AI progress. [1] In this supplement we only consider **AI-relevant compute**. We frequently refer to **H100-equivalents (H100e)** throughout the supplement, which we define as AI-relevant compute with processing performance converted into units of the Nvidia H100 GPU. Processing performance is most directly relevant to training compute, but is also a strong proxy for inference compute too. [2]

In this supplement we model **AI-relevant compute production** in Section 1 by looking at supply chain bottlenecks and estimates that the globally available **AI-relevant compute will grow by a factor of 10x by December 2027** (2.25x per year) relative to March 2025 to 100M H100e.

1. Measured in Floating Point Operations (FLOP), which is a unit of computation referring to the multiplication or addition of floating point numbers of a particular bitlength (in this doc we will assume fp16). AI model training and inference, at the systems level, consist almost entirely of matrix-matrix multiply+add operations of floating point numbers, making FLOP a unit of computational cost.

2. Inference tends to be bottlenecked by memory bandwidth, which acts as a haircut on the total processing performance (TPP) that can be used by the inference workload. Nonetheless, in the section on inference compute for running research agents we perform calculations in terms of memory bandwidth instead of TPP.

1.35x 1.65x

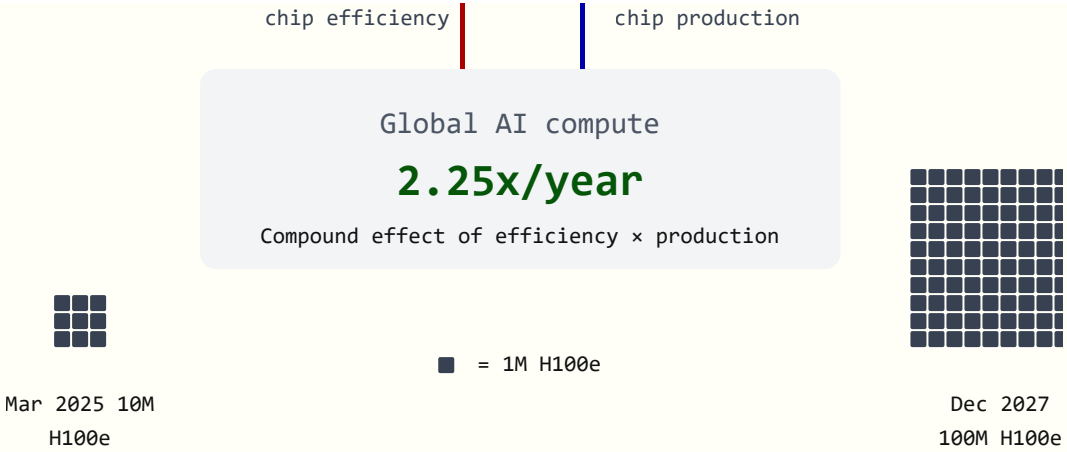


Figure 2: We project the global stock of AI-relevant compute to grow 10x by December 2027.

In Section 2 we model the **distribution of compute** among relevant actors, breaking this down both by **owners** of the compute and **end-users** that either rent or also own the compute. We expect usage to concentrate in the hands of leading AGI companies (e.g., OpenAI, Anthropic, xAI) and the AGI-focused development efforts within larger tech companies (e.g., Google, Meta), the largest two or three of which will have a 15-20% share of the globally available AI compute by the end of 2027 (around 15-20M H100e) up from a 5-10% share today (around 500k H100e).

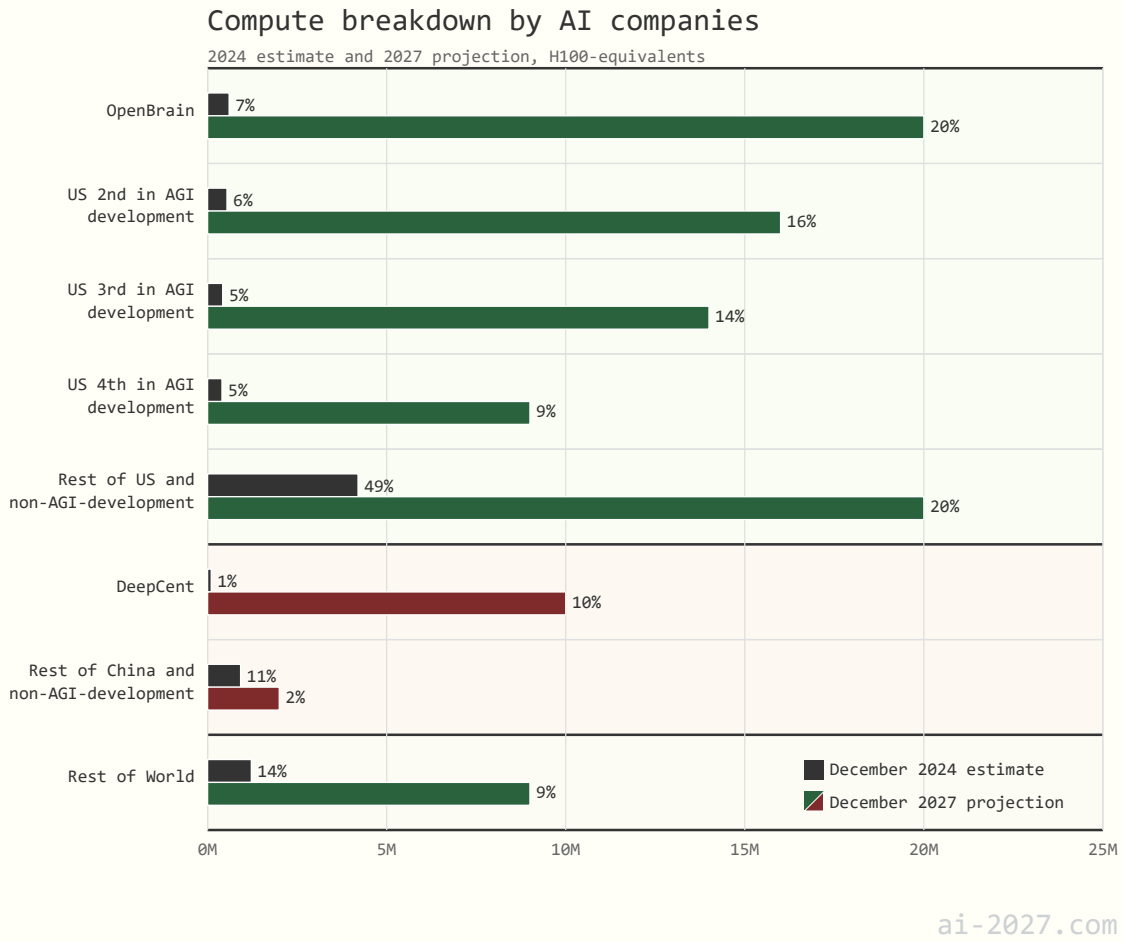


Figure 3: We project the global distribution of AI-relevant compute to concentrate in leading AI companies (with their shares of the global compute stock roughly tripling) and China to maintain a roughly constant share of global compute (around 12%) but to unify it towards a single national AI effort.

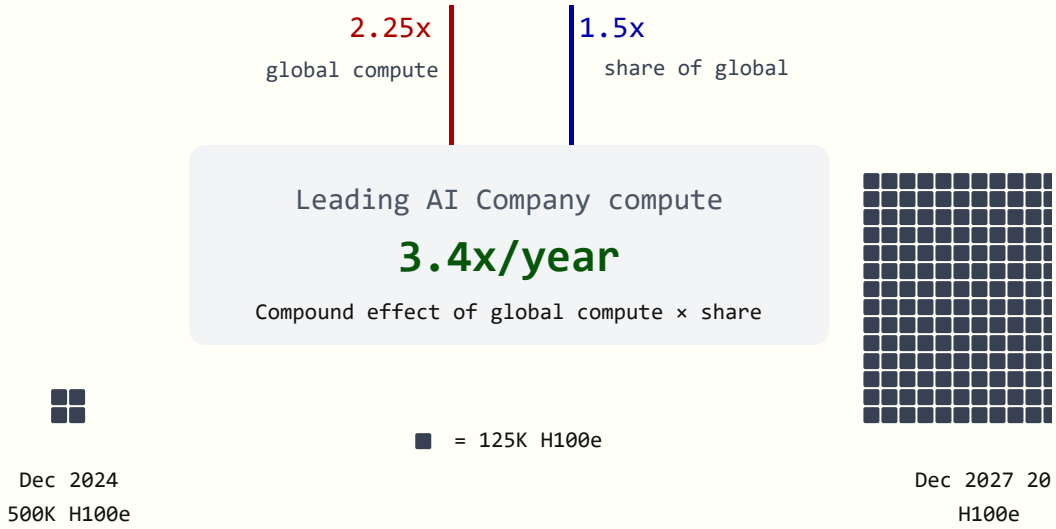


Figure 4: We project the compute available to the leading AI company to grow 40x by December 2027, with a factor of ~10x coming from the global stock of AI-relevant compute growing and ~4x coming from their usage share of the total stock.

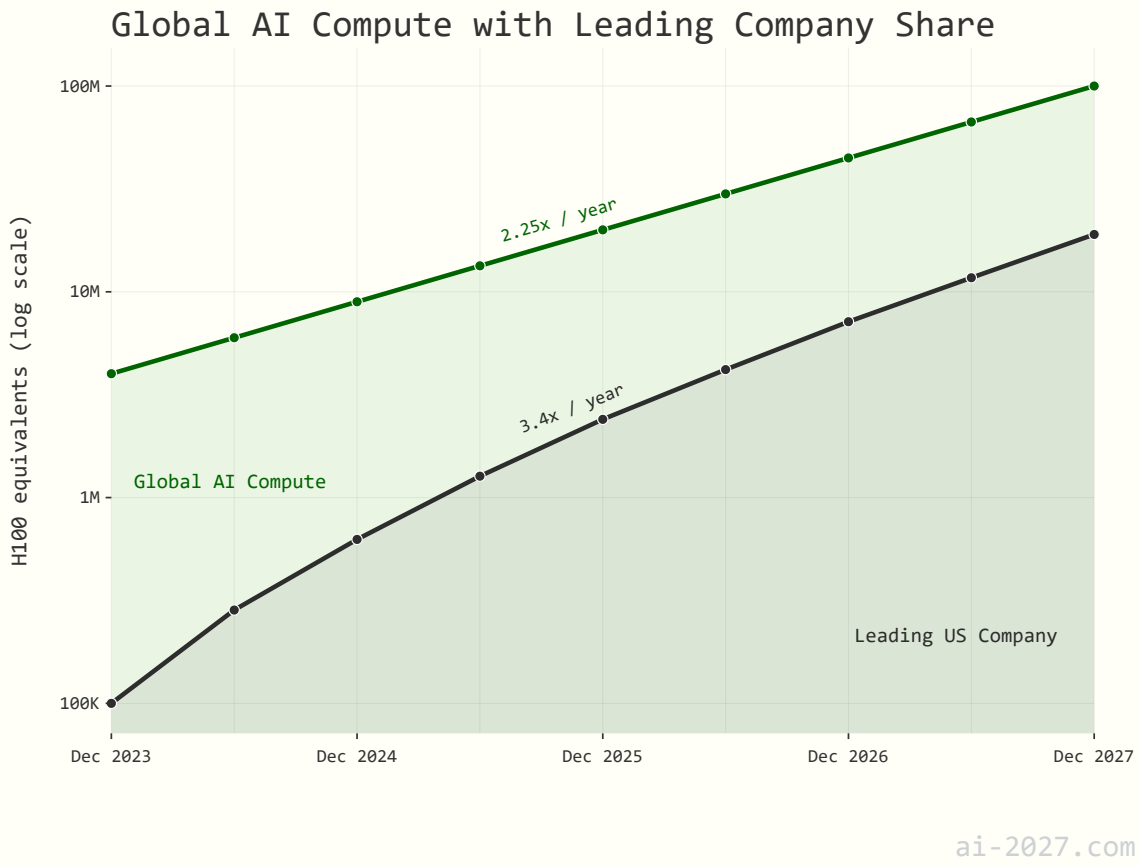
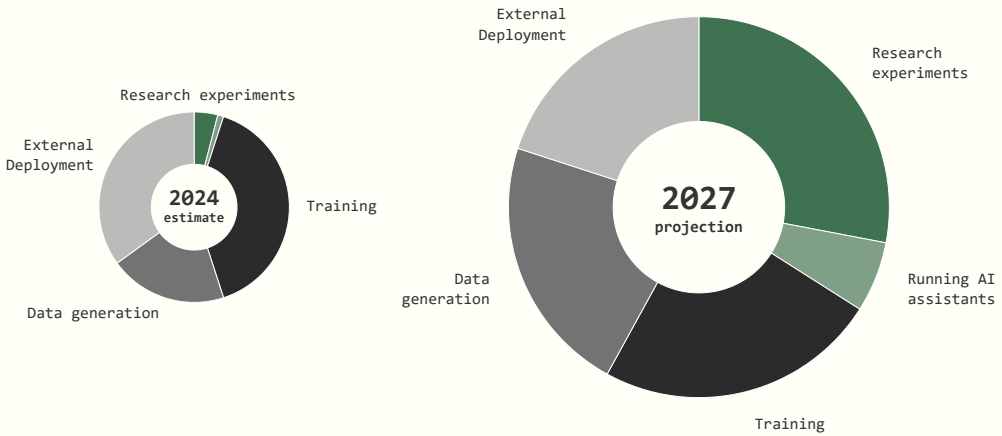


Figure 5: Trend in global compute growth (2.25x/year) and the growing share used by the leading AI company (1.5x/year) through December 2027. The compound effect is a 3.4x/year increase in compute for the leading AI company. FLOP numbers assume 40% model flop utilization.

In **Section 3** we describe how we expect leading AI companies to **use their compute resources**, with shifts away from pretraining and

external deployment, towards post-training, synthetic data generation, and internal deployment (i.e. research automation and research experiments). We project a concentration of compute usage within the leading AI company to research automation, with a relatively small share (5-10%) used on actually running the AIs, and large shares for generating synthetic training data (20%) and giving the AIs large research experiment compute budgets (35%). Actual training runs and external deployment take up smaller shares (20%) but in absolute terms, the compute used on each is still more than 20x greater than in 2024.

OpenBrain's Compute Allocation, 2024 vs 2027



ai-2027.com

Figure 6: Compute use concentrates towards research automation. Data generation also increases.

In Section 4 we do an inference compute analysis for the expected AI research automation in 2027 with estimates on the copies and speed of the AI models deployed. Once they make significant algorithmic efficiency progress by the end of 2027, we expect a leading AI company to be able to deploy about 1M copies of superintelligent AIs at 50x human thinking speed (500 words per second), using 6% of their compute resources, mostly with specialized inference chips.

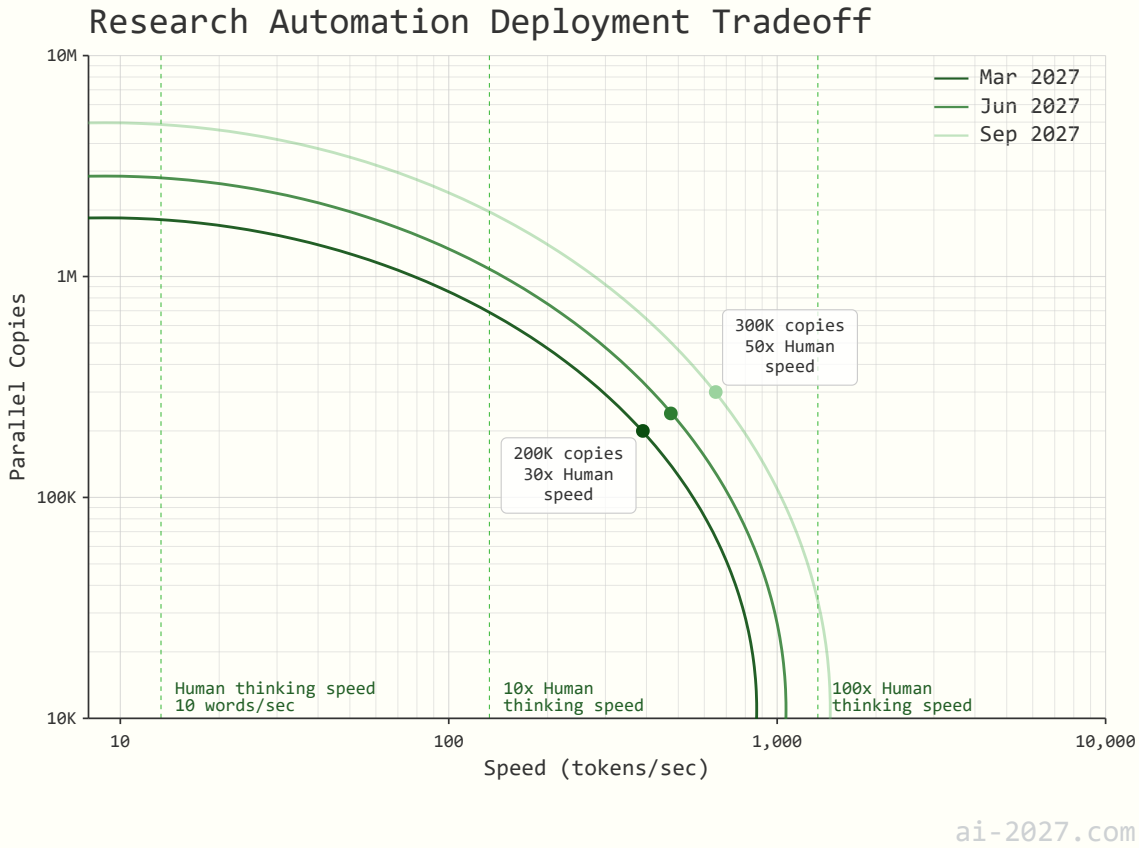


Figure 7: Our expected deployment of AI research assistants for research automation in 2027 using specialized inference chips. The green lines use 6% of the leading AI company's H100e compute.

In Section 5 we look specifically at the training run, total cost, total revenue, and total power usage projections for the projected leading AI company through 2027. We project revenue and cost growth rates to both be around 3x/year, and power usage by the leading AI

company to be around 10GW by December 2027, which implies that they use around 0.8% of the US power capacity, and AI in total uses 60GW globally, 50GW in the US, which is around 3.5% of projected US power capacity (1.35TW, up from 1.28TW today).

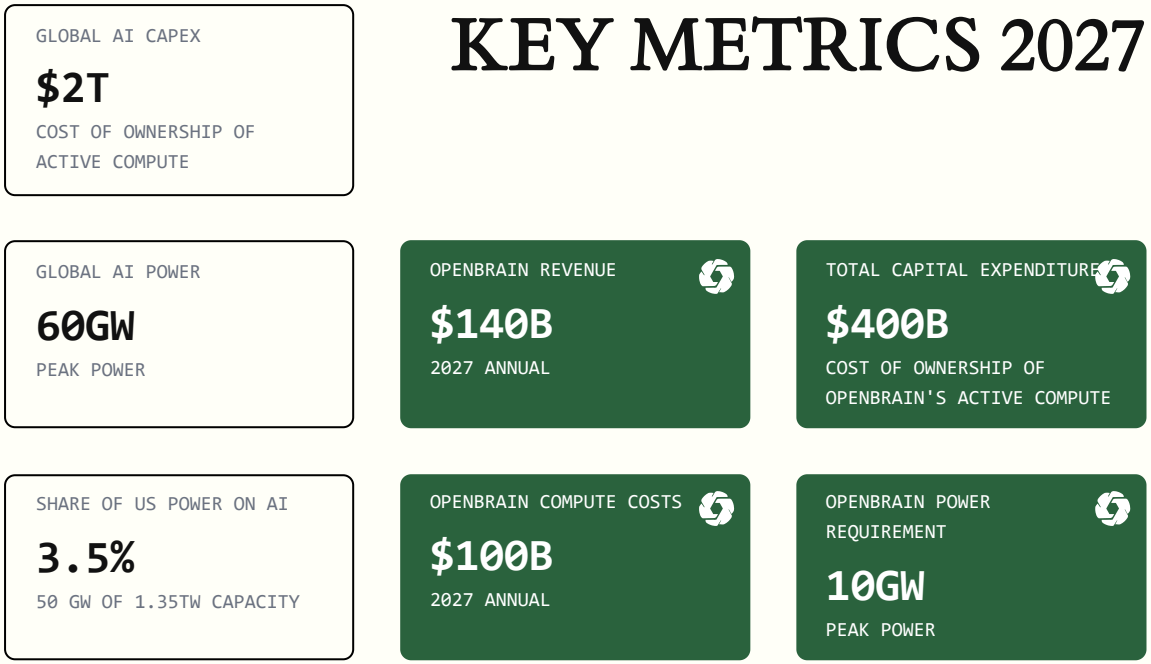
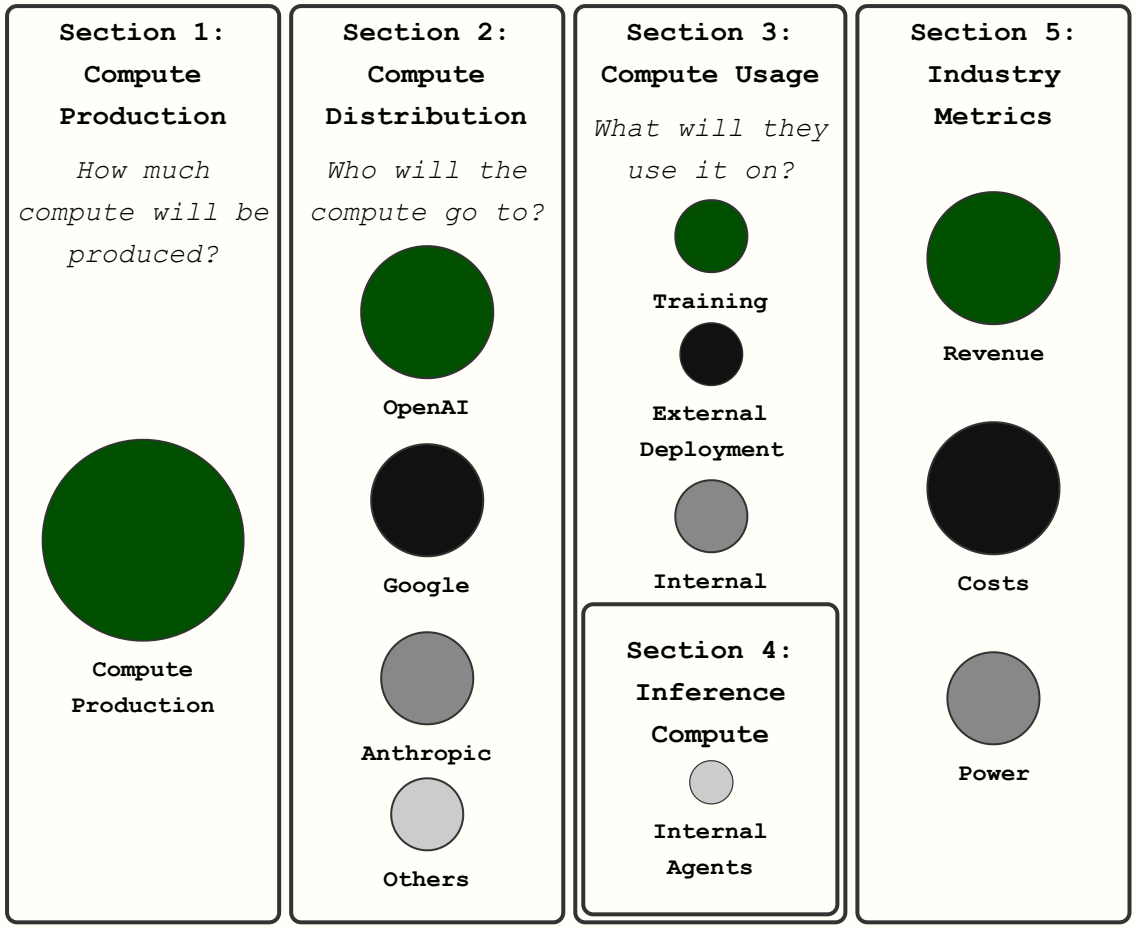


Figure 8: A summary of the projections in Section 5.

All estimates are based on publicly available information which is scarce and uncertain. Broadly speaking, Sections 1 and 2 are

3. Particularly the automated coder milestone being reached in March 2027 as defined and explained in the timelines forecast.

relatively informed and independent of the AI 2027 scenario, while Sections 3, 4 and 5 are far more conditional on the rapid AI capabilities progression that occurs in AI 2027.[3]



Acknowledgements: David Schneider-Joseph, Erich Grunewald, Konstantin Pilz, Lennart Heim, Mauricio Baker, Tao Lin.

4. Measured in Floating Point Operations (FLOP), which is a unit of computation referring to the multiplication or addition of floating point numbers of a particular bitlength (in this doc we will assume fp16). AI model training and inference, at the systems level, consist almost entirely of matrix-matrix multiply+add operations of floating point numbers, making FLOP a unit of computational cost.

5. Matrix multiplications have element-wise independence (so they can be computed in any order).

6. At around 40% model flop utilization (MFU)

7. Inference tends to be bottlenecked by

Introduction

Computing power is one of the key inputs to AI progress. [4] Most computations used in AI training and inference can be completed in parallel, [5] so AI chips or AI accelerators are computational devices (e.g., GPUs) which are particularly efficient at such parallel computation, making them far more effective than traditional processors like CPUs for AI computing workloads.

Indeed, in this supplement we only consider **AI-relevant compute** which we define specifically as any computational unit capable of achieving Total Processing Performance (TPP) of at least 4,000 and Performance Density (PD = TPP / die size) of at least 4. This definition is set just below the A100 SXM GPU, NVIDIA's state of the art chip in 2021. For reference, Nvidia's H100 GPU has a marketed ~15,800 TPP and 19.4 PD. [6] Intuitively you can think of our definition as 'anything at least ¼ as efficient as the H100 counts'.

We frequently use **H100-equivalents (H100e)** as units of AI-relevant compute throughout the supplement, which we define as compute processing performance (in TPP) converted into units of H100 processing power (~15,800 TPP). The choice to focus on TPP makes the analysis herein most directly relevant to training

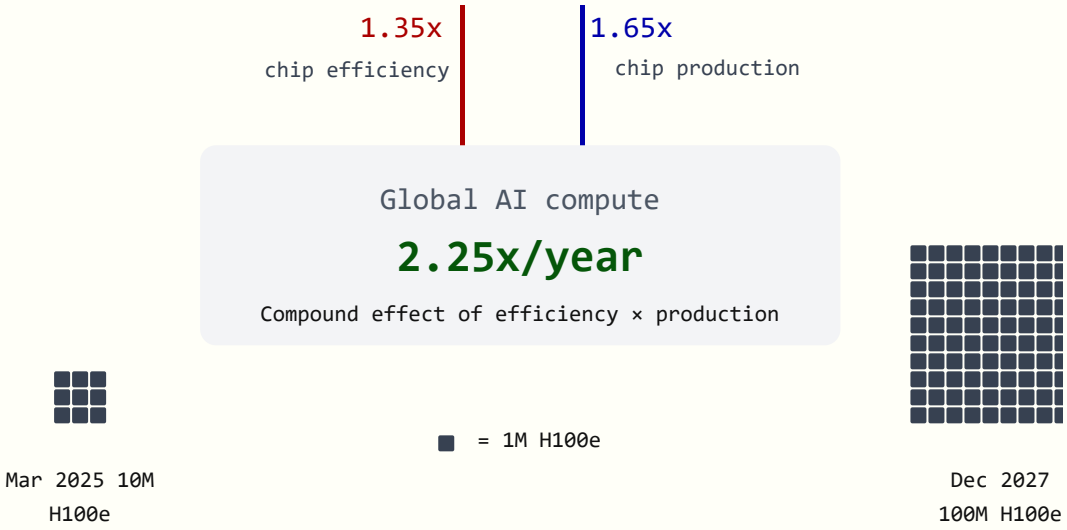
memory bandwidth, which acts as a haircut on the TPP that can be used by the inference workload.

compute, but is also a strong proxy for inference compute too.[7] Nonetheless, in the section on inference compute for running research agents we perform calculations in terms of memory bandwidth instead of TPP.

Section 1. Compute Production

BACK TO NAVIGATION

Status: Uncertain but informed. This section is an informed forecast based on public information.



8. 2027 sees efficiency slightly above trend due to hardware R&D automation.

9. Assumes chips are usable for roughly 4 years.

10. Assumes roughly 2M H100e were available prior to 2023.

11. Includes all computing hardware (servers, networking, etc.), excludes datacenter and intra-campus infrastructure (cooling, power, interconnection, etc.).

12. Includes all computing hardware (servers, networking, etc.), and datacenter infrastructure (cooling, energy, etc.).

13. Assumes Epoch AI's 1.26x per year power efficiency trend. The H100 to B100 achieved 2.5x FLOP performance with 1.7x power consumption, implying a 1.2x per year improvement in power efficiency. We expect the jump to the Rubin to be even better proportionally, given the change in underlying node from N4 to N3.

14. This can be calculated by downloading

Figure 9: Summary of our compute production forecast.

We expect the total stock of AI-relevant compute in the world will grow **2.25x per year** over the next three years, from **10M H100e today to 100M H100e by the end of 2027**. We estimate this by decomposing growth in total compute availability, measured in H100e, into improvements in (A) chip efficiency and (B) chip production, contributing a baseline growth rates of 1.35x and 1.65x respectively.

	2023	2024	2025	2026	2027
Performance density (PD) multiplier on average chips produced (H100 = 1x)	.66x	.9x	1.22x	1.64x	2.4x [8]
Total AI chip area produced (H100-sized chip = 1)	3M	5.5M	9M	16M	25M
Total AI chips produced in H100e processing performance (TPP)	2M	5M	11M	25M	60M
Cumulative H100e available [9]	**4M** [10]	**8.5M**	**18M**	**40M**	**100M**
Total cost of ownership per H100e [11]	$50k	$40k	$25k	$20k	$15k
Total AI datacenter spending [12]	$110B	$270B	$400B	$600B	$1T
Total datacenter power requirement per H100e [13]	1.3kW	1.0kW	750W	700W	550W
Total AI datacenter power requirement	5GW	9GW	15GW	29GW	62GW

Epoch's historical data and dividing the FP16 and FP32 performance by the die area of the ML hardware data, and plotting the trend. Originally I (Romeo) had eyeballed the ~flat die size trend and chip performance trends and assumed it was also 1.35x/yr, and forgotten to look into it precisely. As of December 2025, thanks to Robi Rahman pointing this out in this X post I finally checked it out precisely, and found it was actually also exactly 1.35x/yr for the most relevant/longstanding precision formats. More explanation and graphs in this X reply.

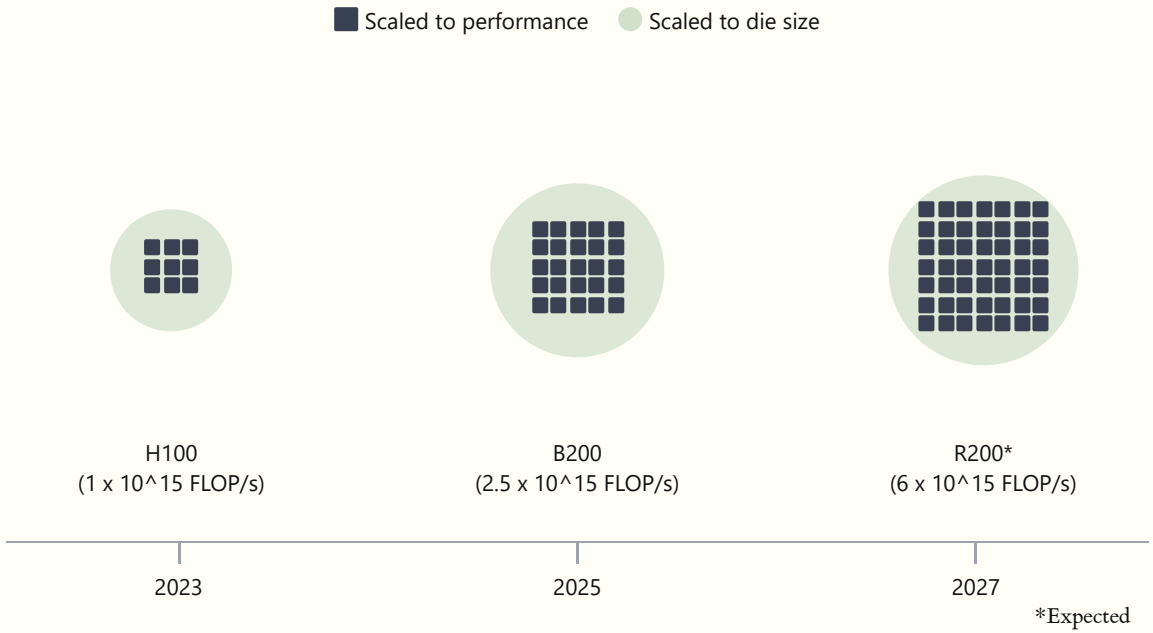
13. Nvidia have since put out their GTC 2025 announcement which has given more concrete details on the Rubin GPU series, but we have not yet incorporated this news into the forecast.

16. Likely mostly of the gain will be from increased transistor density.

See also the full spreadsheet model.

Chip efficiency

For improvements in chip efficiency, (performance / area) we extrapolate Epoch AI's historical trend of **1.35x per year**, [14] and find it is consistent with the already reported performance of upcoming chips such as NVIDIA's GB200 as well as rumoured plans for the Rubin series detailed below. [15]



17. We obtain the 1.1x estimate by observing that the B200 achieved 2.5x performance increase over the H100 despite only a 2x increase in die size on the same TSMC N4 class node, meaning that a 2.5/2 = 1.25x improvement was achieved through other methods, and adjust down given likely reduction of this within-node optimization given the N4 to N3 transition.

18. 2x increase from the H100 to the B200 and a further 1.2x increase from the B200 to the R100.

19. 6 / 2.4 = 2.5

The widely adopted state of the art GPU in 2024 is NVIDIA's H100, which has 1e15 FP16 FLOP in raw performance. In three years, we predict the widely adopted state of the art GPU to be NVIDIA's Rubin GPU (R200), which we project to achieve a ~2.4x improvement over the B200 (widely used in 2025-2026) to 6e15 FP16 FLOP performance. We think this will be achieved through a 1.2x increase in die size, a 1.7x increase given the transition from TSMC's N4 to N3 process, [16] and a 1.1x increase from other miscellaneous improvements. [17]

To avoid confounding efficiency with the increase in die size, we need to adjust this overall 6x increase in the SOTA GPU performance from the H100 to the R200 down for the roughly 2.4x increase in die size between these chips. [18] This means we get an overall 2.5x increase in chip efficiency over the next three years, [19] matching the 1.35x per year historical trend. We assume that the trend on frontier NVIDIA GPUs is a good proxy for the trend on the average GPUs available each year, not only because they will make up most of the chips, but also because the performance of other popular chips (such as Google's TPU series) should rely on highly correlated upstream factors such as TSMC's manufacturing process advances.

Chip production

20. Of which 650k were actual H100s, with the rest being mainly A100s, or the custom made A800s, and H800s for the Chinese market.

21. This source implies about 3M H100e were shipped by Nvidia (~1.8M among the top 10 buyers making up 60% of the total). Looking at Google TPUs, Amazon Trainium and AMD chips and other in-house chips by Meta and Microsoft

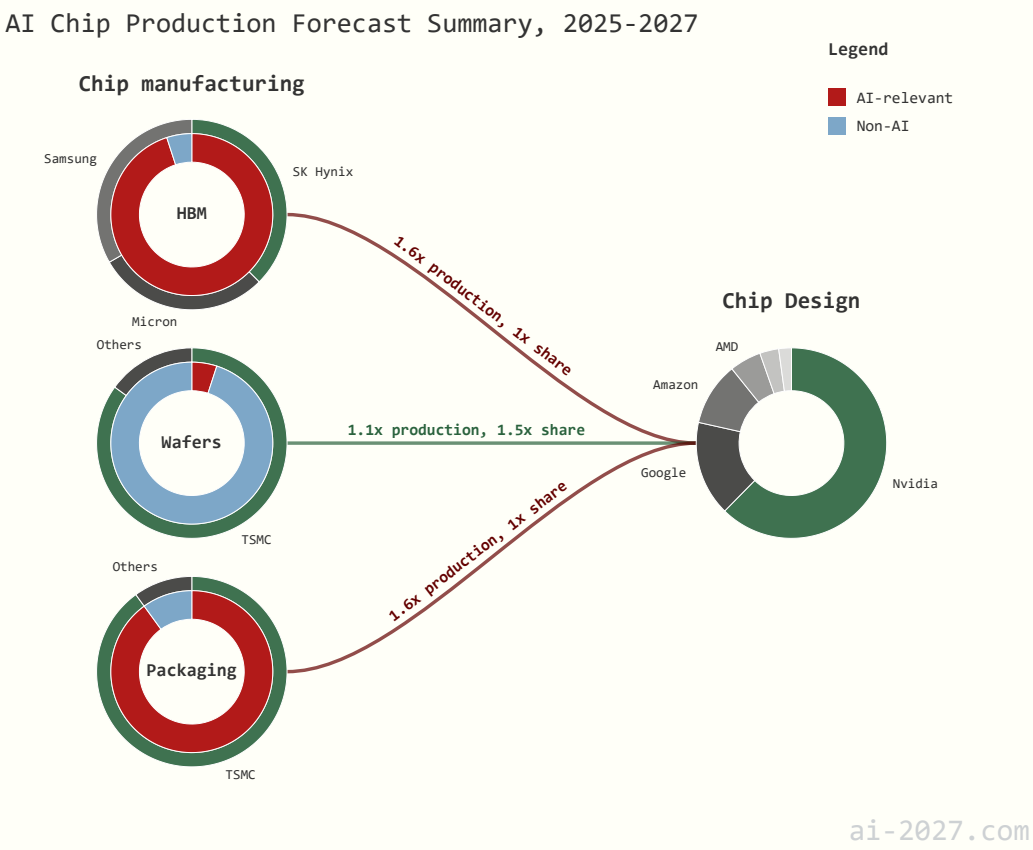


Figure 10: Summary of our chip production model.

For increases in chip production we further decompose into three key components driving production: (B.1) wafer production, mainly fulfilled by TSMC's N5 and N3 processes, (B.2) advanced packaging capacity, also mainly serviced by TSMC's CoWoS technology, and (B.3) high bandwidth memory (HBM), mainly supplied by SK Hynix, and increasingly so by Micron and Samsung.

contribute another estimated 2M H100e. This roughly matches up with the ratio of global spending of $230b on AI infrastructure (of which around 80% or $180b is likely on actual chips) with Nvidia's $100b chip revenue.

22. There were about 2.5M 12-inch N3 and N5 wafers produced in 2023 by TSMC (which we backed out by looking at their revenue by node and their node pricing), which could each produce about 28 functioning H100-sized chips. Google's TPUs are less performant but also smaller, but older NVIDIA chips would use a larger than proportionate wafer area, so we assume 2M total H100-sized chips shipped. Overall, there were 2.5M * 28 = 70M H100-sized chips of wafer capacity then AI accelerators used a mere 3% of TSMC's combined N3 and N5 capacity.

We estimate 1.2M H100e to have been shipped in 2023 by Nvidia [20] and for these to have made up about 60% of the total AI compute market, mostly due to Google's in-house TPU production supported by Broadcom, for a total of **2M H100e in 2023**. We expect this to have increased to **5M H100e in 2024**, again with roughly 60% coming from Nvidia. [21]

We mainly focus on production numbers for TSMC, and SK Hynix since they make up about 90% and 60% of their respective market shares, and expect trends in the rest of the market to be similar. Overall, over the next three years, we project **AI chip production to be bottlenecked by advanced packaging and HBM production to about 1.65x per year**.

Wafer Production

TSMC's 4N node (within their N5 process) has been chosen for Nvidia's next generation of Blackwell GPUs, while the N3 process will likely be used for the Rubin GPUs projected to enter mass production in late 2026. In 2023 AI accelerators likely used at most 3% of TSMC's combined N5 and N3 production lines, [22] and given reports of these production lines sometimes running at as little as 70% capacity, wafer production is very unlikely to bottleneck AI chip production over the next three years. Even if production

doubles each year, AI chip production shouldn't reach more than ~40% of the fabrication capacity by 2027, even ignoring new production capacity coming online (at a rate of ~15% expansion per year). Notably, beyond 2027, we'd expect wafer production limits to slow the growth rate of chip production from 1.65x to around 1.25x.

Advanced Packaging

AI accelerators require advanced packaging to create dense connections between logic and memory for the high throughput required by AI workloads. Currently TSMC's advanced packaging capacity is mostly used by AI accelerators and has been reported to have expanded 2.5x from 2023 to 2024. Given TSMC's expectation of 50% annual growth rate in AI demand, we don't expect them to expand these production lines too aggressively, in fact they have announced plans to increase capacity by 1.6x per year through 2026. Though given the ~2.5x expansion seen in 2024, we expect raw production expansion to beat expectations and continue at around 2x/year. Though at the same time advancing chip efficiency will require the difficulty of manufacturing processes required to also beat expectations (specifically moving to 3D packaging or future advances). Therefore, while we model the raw production capacity grow at 2x/year (ignoring increased difficulty), we then adjust

down by 1.2x/year for the manufacturing difficulty (e.g., due to lower yields,changing production lines to meet new requirements) for an overall rate of 1.65x per year over the next 3 years.

High Bandwidth Memory

AI workloads are memory intensive and require expensive, fast High Bandwidth Memory (HBM). HBM production is an advanced process that involves precisely stacking and connecting several DRAM chips. SK Hynix were the first to develop this technology, and are the first to be producing the latest generation HBM3e. Micron and Samsung are also catching up with roadmaps to be competitive with SK Hynix for future HBM4 generations and beyond. HBM production lines have been reported to be expanded 2.5x in 2024 compared to 2023, but SK Hynix are only projecting a 1.6x increase in demand going forwards. Similar to advanced packaging, we expect HBM production (ignoring difficulty) to beat these expectations and grow at 2x, but again adjust down by 1.2x given the increasing manufacturing difficulty (in particular the move from 8-Hi, to 12-Hi and 16-Hi stacks) to give a rate of 1.65x per year over the next three years.

Hardware R&D automation

23. The taxonomies are trivially exhaustive because they both include a 'rest of the

In line with our broader capabilities and AI R&D projections, we expect leading AI companies in 2027 to automate specialized chip designs and increase in-house chip production. There are early signs of such ambitions, such as OpenAI's hiring of prolific Google TPU designers and plans to design their first in-house chip in 2025. Though as 1.35x per year improvements in chip efficiency keep getting harder and harder to achieve, and our expectation that research automation will mostly be directed elsewhere, we expect most of these effects to provide a marginal increase in effect on the average chip efficiency to 2.4x the H100 (rather than 2.2x which would match the baseline trend), mostly through a wave of production of inference specialized chips.

world' category, and are designed to be most illustrative of who we believe will be the most relevant actors.

24. We expect the leading AI companies such as OpenAI and Anthropic to start owning large amounts of their own compute by 2027. In this section we just model a single leading company given that we expect a lead in AI development to have compounding effects (not only from revenue leading to more compute access but also from internal research automation). This is a relatively high point of uncertainty and the 'leading AI company' projections herein can really be viewed as the top 1-3 AI companies.

Section 2. Compute Distributio

BACK TO NAVIGATION

Status: Uncertain but informed. This section should be read as an informed forecast based on limited public information.

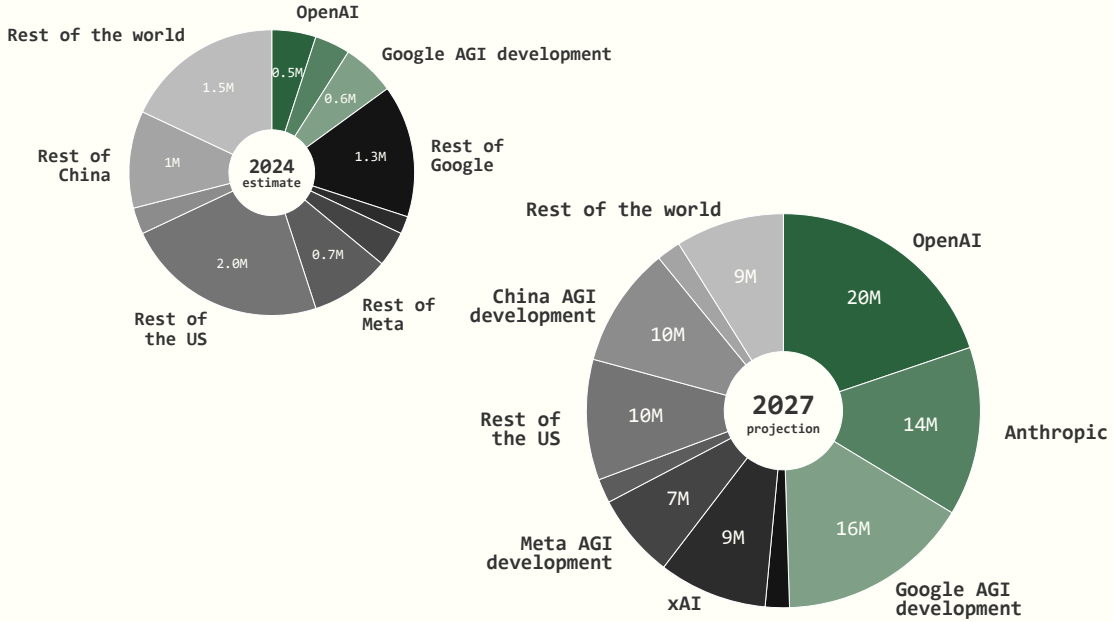


Figure 11: Summary of our compute distribution projection to the end users of the compute.

In this section we project the share of the world's AI-relevant compute owned by compute providers and the share of AI compute utilized by end-users, through 2027. **Compute owners** are entities that own and operate AI compute clusters. **End-users** are entities that use the compute clusters for their AI workloads. In some cases,

entities are both providers and end-users (e.g., Google, Meta, xAI). In other cases, owners mostly rent out their compute to other entities (e.g., Microsoft renting to OpenAI, Amazon renting to Anthropic, Oracle renting to OpenAI). We use the following exhaustive taxonomy of owners and end-users. [23]

Taxonomy of relevant actors

Owners	End-users
Microsoft	OpenAI
Google	Anthropic
Amazon	Google AGI development
Meta	Rest of Google
xAI	Meta AGI development
Oracle	Rest of Meta
China Big Four (Bytedance, Alibaba, Tencent, Baidu)	xAI
Leading AI Company [24]	China AGI development
Rest of the US	Rest of the US
Rest of the world	Rest of the world

According to our projections, **leading AI companies will have 20-40x as much AI compute by December 2027 in absolute terms,**

compared to December 2024, with a factor of 10x coming from the previously covered growth in compute availability, and a factor of 3x coming from increased concentration of compute in leading AI companies and their AGI development efforts.

Compute owners

Recent past

In the compute production section we estimated there being 4M H100e of AI-relevant compute available globally at the end of 2023, and 8.5M by the end of 2024. Previous work by Epoch AI in Appendix B of *Can scaling continue through 2030?* estimates that Meta and Microsoft each had 150,000-600,000 H100e at the start of 2024, and that Google and Amazon each had 400,000-1.4M H100e. Our point estimates for the start of 2024 (December 31st 2023) are uncertain given the scarcity of public information, but they fall in the middle of Epoch AI's ranges and align with several overlapping sources. We use this report of 2024 production and purchases to produce estimates for the resulting December 2024 distribution.

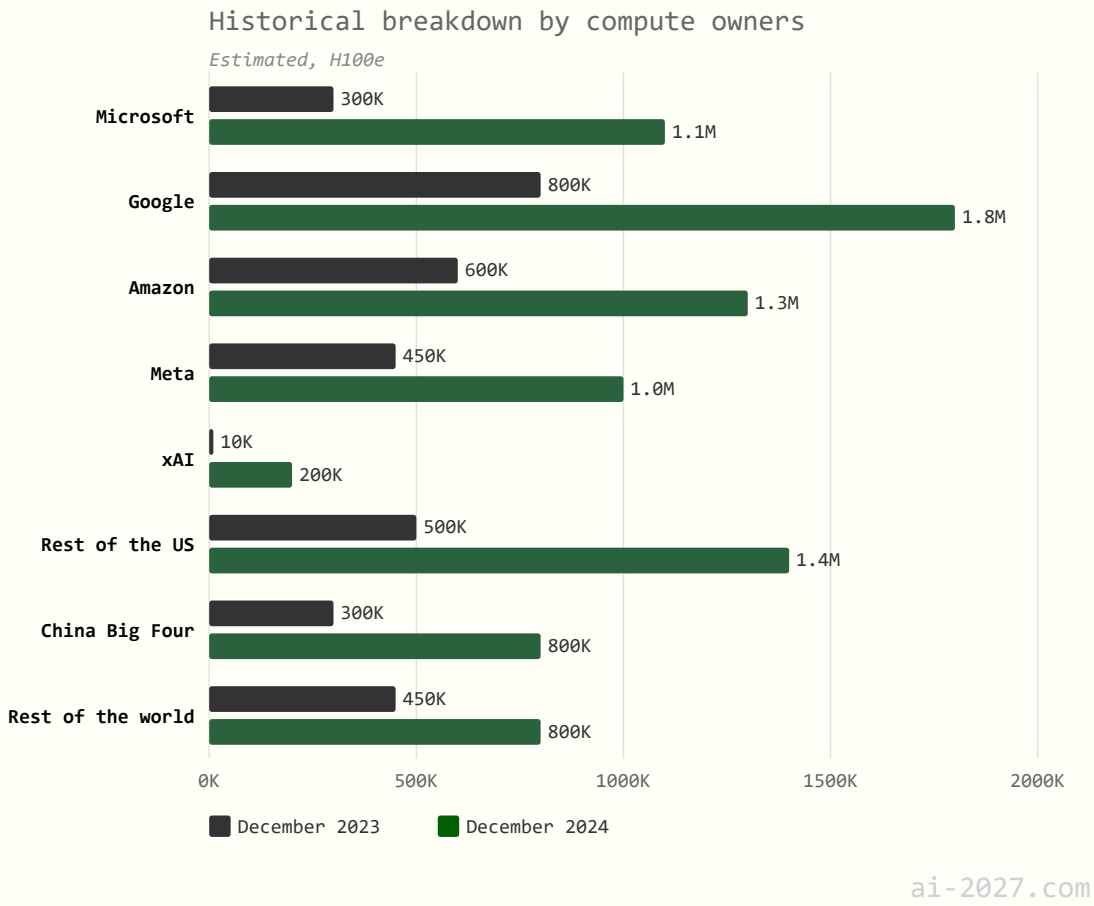


Figure 12: Estimated historical AI compute breakdown in H100e among owners of the compute.

The top 5 US spenders on AI servers according to Omdia (and reported by the Financial Times) in 2024 are shown below, with our estimate of how many H100e they acquired and therefore their spending per H100e. Google's strong spending effectiveness is due

to their in-house TPUs, already in their 6th generation. Amazon and Meta also have significant in-house designs but they are in earlier stages of the R&D process. Microsoft and xAI bought almost entirely Nvidia chips.

	2024 spending on AI servers	H100e gain in 2024 vs. 2023	Spending per H100e
Microsoft	$32B	800K	$39k
Amazon	$26B	700K	$37k
Google	$22B	1M	$22k
Meta	$20B	550K	$36k
xAI	$7B	200K	$37k

Intermediate 2025 Projection

	Projected 2025 spending on AI servers	H100e gain in 2025 vs. 2024	2025 Spending per H100e
Microsoft	$56B	2.4M	$23k
Amazon	$48B	2.2M	$22k
Google	$44B	2.2M	$20k
Meta	$35B	1.6M	$22k
xAI	$30B	1.3M	$23k

For forecasting the 2025 compute distribution we look at the top 5 US companies by spending on AI and their 2024 share of spending on AI servers. We find the following:

- **Microsoft** has announced $80B in AI spending for 2025, and last year 55% of their spending went to AI servers ($31B out of $56B). We expect the share on AI servers to increase to 70%, producing an estimate of **$56B spending on AI servers in 2025.**
- **Google** has announced $75B in AI spending for 2025, and last year 42% of their spending was on AI servers ($22B out of $52B). We expect the share on AI servers to increase to 58%, producing an estimate of **$44B spending on AI servers in 2025.**
- **Amazon** has announced $100B in spending for 2025, and last year 33% of their spending went to AI servers ($26B out of $78B). We expect the share on AI servers to increase to 48%, producing an estimate of **$48B spending on AI servers in 2025.**
- **Meta** has announced $65B in spending for 2025, and last year 50% of their spending went to AI servers ($20B out of $40B). We expect the share on AI servers to increase to 54%, producing an estimate of **$35B spending on AI servers in 2025.**
- **xAI** announced (minute 1:05:00) around **$30B spending on AI servers in 2025.**

25. The buildouts detailed here indicate slightly more aggressive plans for Microsoft compared to Google (despite their current advantage).

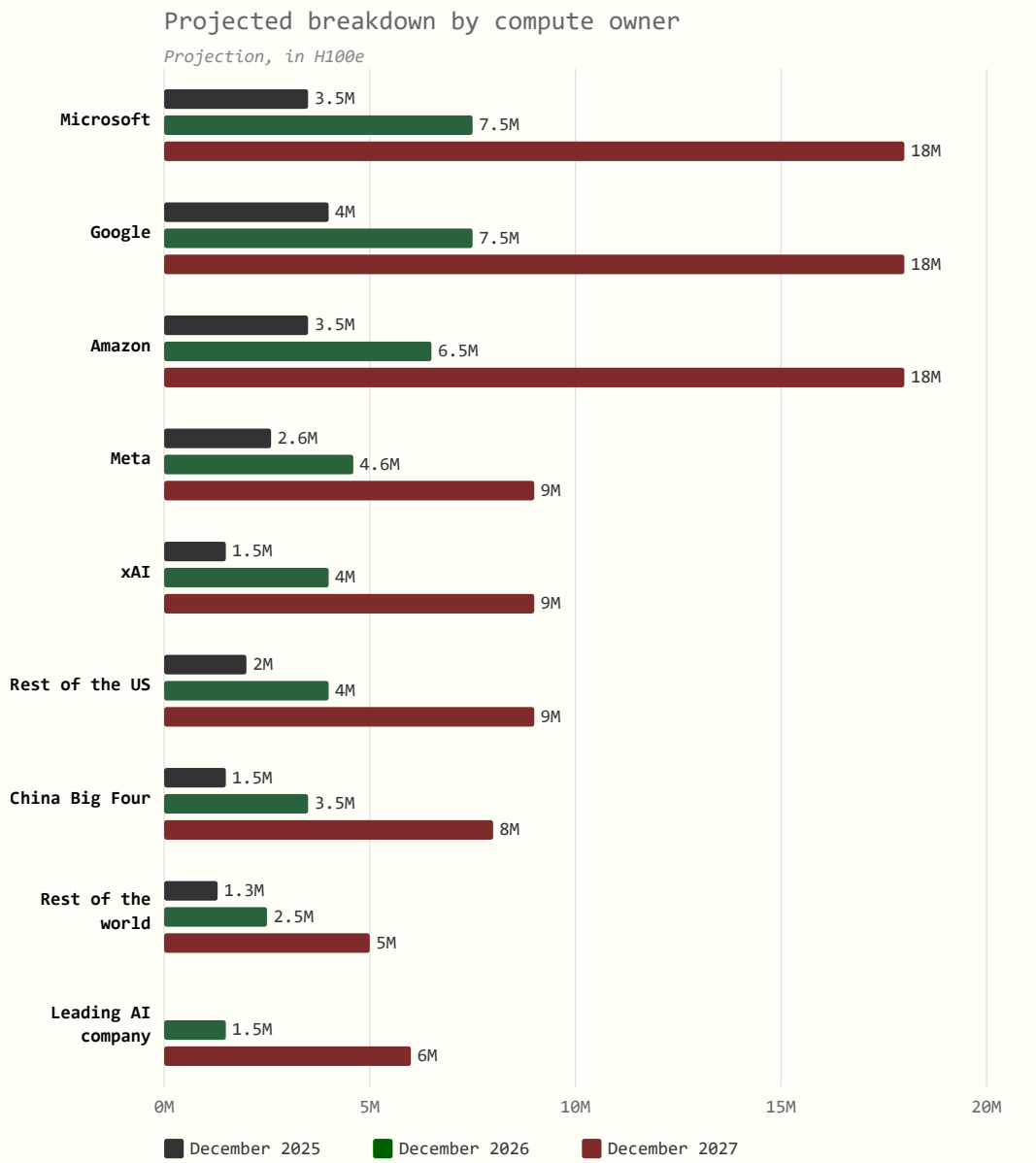

Projected breakdown by compute owner
Projection, in H100e
Microsoft
3.5M
7.5M
18M
Google
4M
7.5M
18M
Amazon
3.5M
6.5M
18M
Meta
2.6M
4.6M
9M
xAI
1.5M
4M
9M
Rest of the US
2M
4M
9M
China Big Four
1.5M
3.5M
8M
Rest of the world
1.3M
2.5M
5M
Leading AI company
1.5M
6M
0M
5M
10M
15M
20M
December 2025
December 2026
December 2027
ai-2027.com

Figure 13: Projected AI compute breakdown in H100e among owners of the compute.

2027 Projection

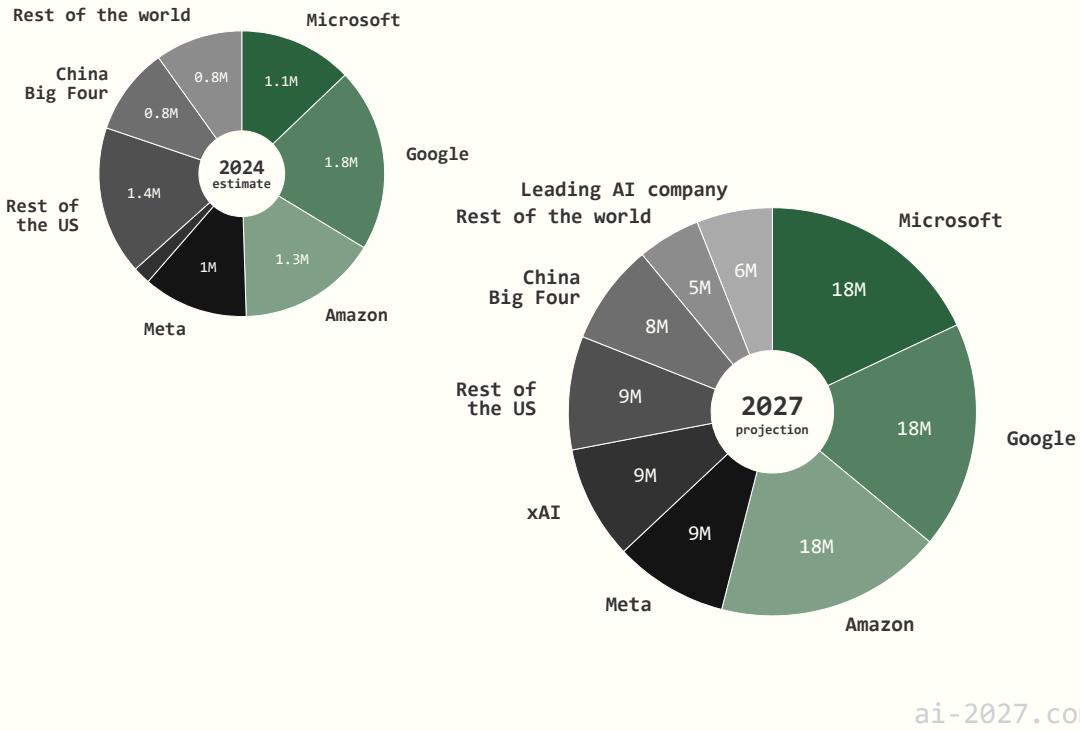


Figure 14: Projected AI compute owner shares, values in H100e.

By 2027, our compute production section has the total compute available growing to 100M H100e which corresponds to a 2.25x

increase per year. We should expect this explosive level of growth to easily shake up the distribution of compute. We expect this growth to be driven by sustained trends in demand from end-user AI developers that start to have annual revenues in the tens of billions from their AI models.

We make the following projections:

- **Microsoft** (driven by their relationship to OpenAI) scale aggressively, going from **13% to 18%** of the world's compute share.[25]
- **Google's** existing compute overhang, leads to a lower rate of expansion and their compute share edges down from **21% to 18%**.
- **Amazon**, scales aggressively with sustained demand from Anthropic and their own AI efforts justifying aggressive buildouts that inches their share up **to 18%**.
- **Meta's 12%** share slips to **9%** because revenues from their mostly open source AI models lag behind and don't justify the same degree of aggressive buildouts in 2026 and 2027. Note this still means their absolute compute increases 7x from 1.1M to 7.5M H100e.
- **xAI** scales aggressively, with total hardware capital expenditure reaching around $100B by 2027, backed by decent AI revenue

and Elon Musk's backing, growing their compute share from **2% to 9%**.

- The **rest of the US**, which includes larger players like Oracle and Coreweave as well as smaller providers and owners, get somewhat crowded out and have slightly lower economic incentives to scale as aggressively as the bigger players. Overall, the rest of the US's share falls from **18% to 9%**.
- **China's** Big 4 start to face marginally more difficulty in securing AI compute due to export controls, although China's domestic production ramps up and produces chips in large quantities at a quality around 3 years behind Nvidia. Nonetheless their share only dips from **9% to 8%** since US export controls stay poorly enforced, smuggling in significant quantities to other Chinese entities means the total compute in China is significantly higher. Domestic production also ramps up through SMIC but is years behind in efficiency.
- The **rest of the world** also gets somewhat crowded out for similar reasons to the smaller US players, and their share dips from **9% to 5%**.
- Finally, we expect a future **'leading AI company'** to decide to build their own datacenters in the latter end of 2026 and during 2027. We believe this is consistent with their revenues reaching an annualized run rate of $50B by the end of 2026, and their AI

capabilities helping them develop specialized in-house GPUs in 2027. Overall, we predict that with about $90B of spending they will efficiently acquire around **6%** of the world's AI compute (excludes the R&D costs of in-house chip designs).

Compute end-users

Recent past

- **OpenAI's** recent cost reports indicate that they are renting an average of around 250k H100s through 2024, which we expect to be tail-heavy reaching 460k by the end of the year, mainly supplied directly by Microsoft, but also through deals with Oracle and Coreweave. This puts their share of the global 8.5M H100e at around **5%** by the end of the year.
- **Anthropic's** compute usage is expected to be slightly behind at **4%**, given lower revenues, users, and therefore access to capital. We expect them to have access to around 360k H100e by the end of the year, a lot of this being recently added by a 400k Trainium2 cluster provided by Amazon.
- **xAI** has an expanding 200k H100 cluster, putting their share at **2%**.
- **Google and Meta's** AI compute fleets make up **21%, 13%** of the world's share, but we only expect them to be using around 30%

of these clusters for AGI development. Most of the rest of Google and Meta's AI compute is dedicated to developing and servicing their large suite of recommendation algorithms around the world.

- We expect most of the cloud service demand from the rest of Amazon, Microsoft, Neocloud and smaller providers fleets to be used by companies and startups throughout the **rest of the US** and **rest of the world**, each with shares of around **20%**.
- We expect **China** to use essentially all of the compute owned by China's Big 4, as well as around half of what is owned by the 'rest of the world' category, given how much of this is concentrated in Malaysia and Singapore, which is likely used by (or even directly smuggled into) China. This leads to an estimate for a total **14%** compute usage, of which we expect 20% to be dedicated to AGI development in late 2024 for a total China AGI development share of 3%.

2027 Projection

*For the sake of this section, we assume that OpenAI will be the leading AI company in 2027. The projections here are therefore just an illustration of how the compute breakdown would look under a scenario where OpenAI remains the leading AI company.***

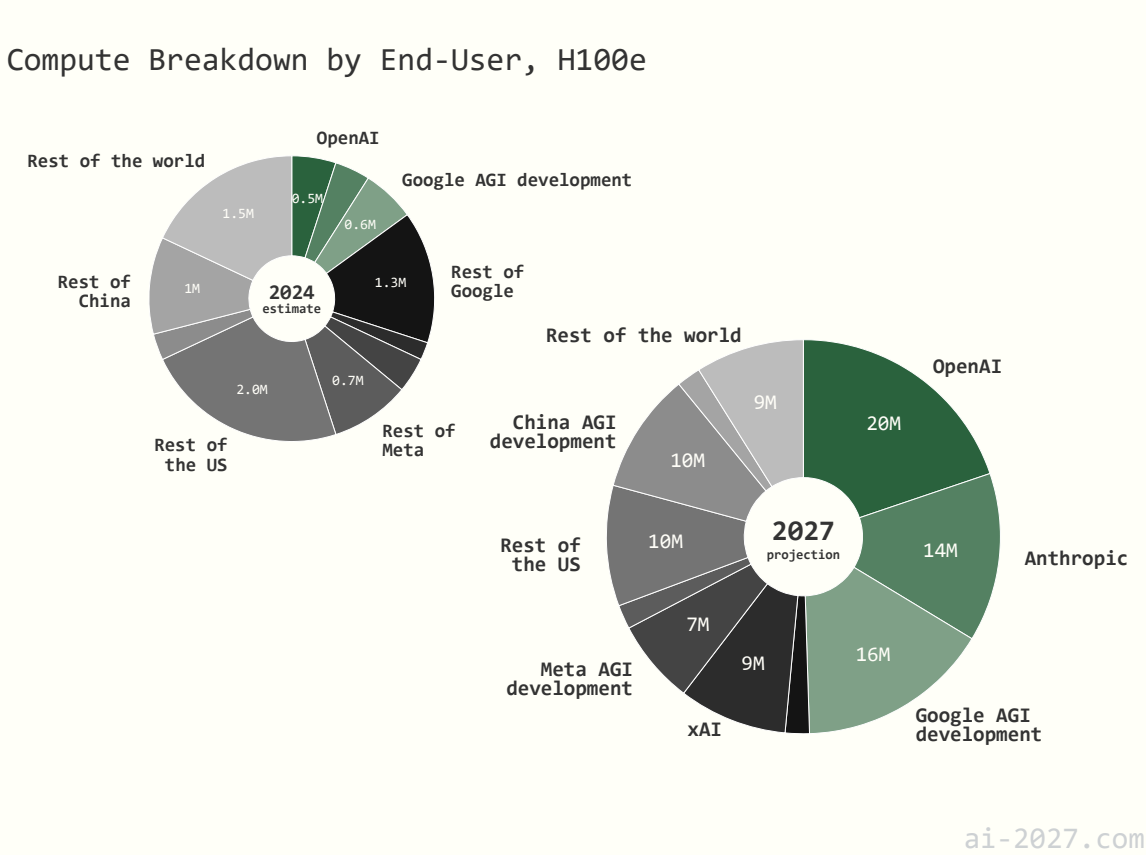


Figure 15: Projected AI compute end user shares, values in H100e.

- **OpenAI and Anthopic** both scale aggressively, motivating aggressive buildouts by their investors Microsoft and Amazon, eventually owning some of their own datacenters. **OpenAI's** usage share of the world's AI compute jumps from **5%** at the end of 2024 **to 20%** and **Anthropic from 4% to 14%**.

26. As measured by hourly rental cost.

27. In the slowdown ending this spikes.

- xAI continues to use the entirety of their owned compute explained in the compute owner section, so their usage share also goes from **2% to 9%**.
- **Google and Meta** now dedicate 90% and 80% respectively of their compute to AGI development so Google's AGI development share grows from **6% to 16%**, and Meta's AGI development share from **4% to 6%**.
- **China's** total compute share stays at around 13%, but now they dedicate 90% to AGI development as opposed to 40%, so the China AGI development share goes from **2% to 12%**.
- Finally, for similar reasons to the providers section, we expect the **rest of the US and rest of the world's** end users to be crowded out, and their usage shares to dip from **23% to 11%** and **18% to 11%** respectively.

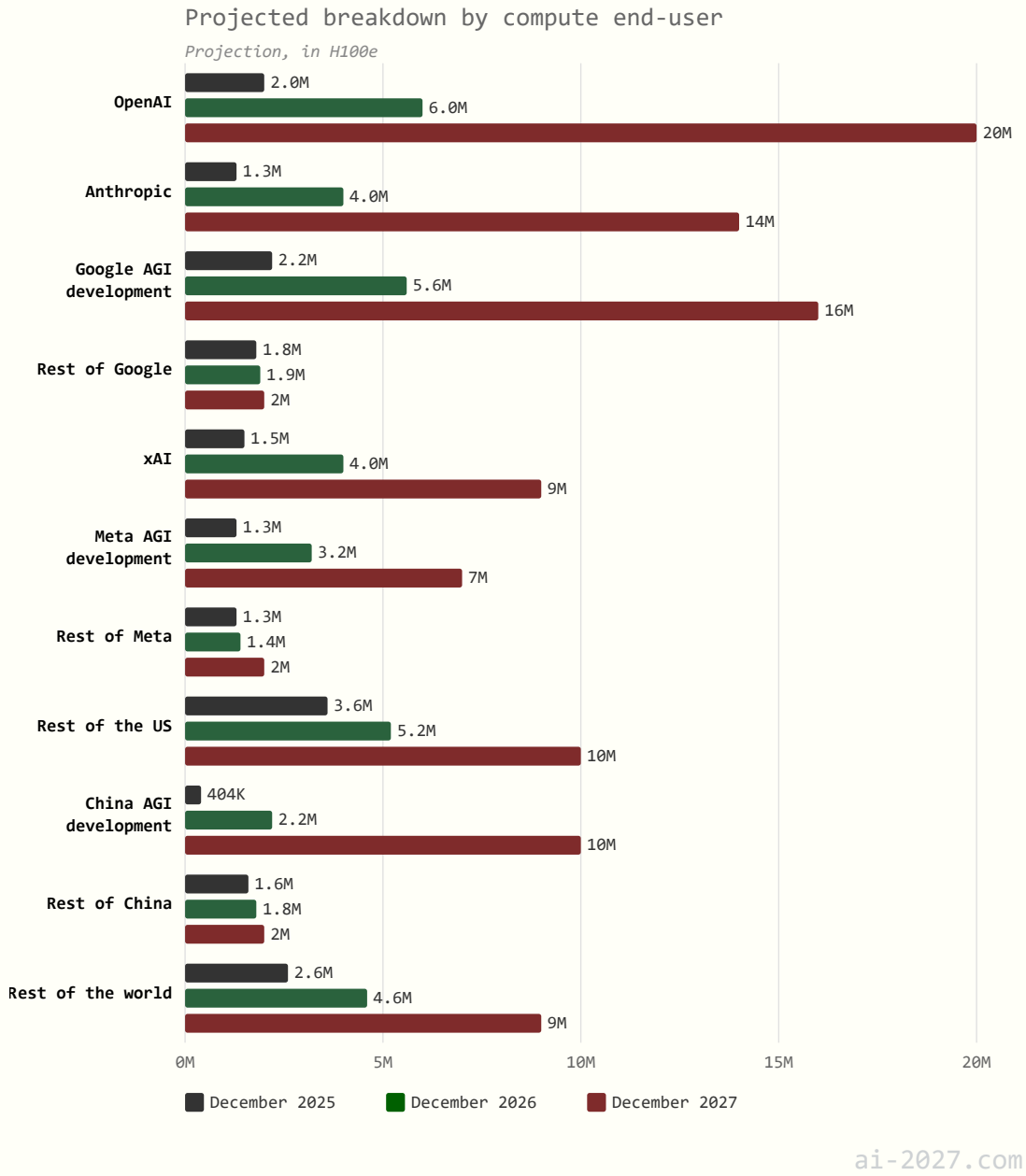


Figure 16: Projected AI compute end user breakdown in H100e.

28. Adjusting for an estimate of 35% model flop utilization on pre-training and 20% during post-training workloads.

29. Though we do expect them to be pushing the boundaries with clever training algorithms to distribute their largest training runs, potentially across 2-3 datacenters.

30. At this point we expect absolute training compute to stay constant, so the share decreases as post-training workloads are not modified (partially due to security concerns around moving model weights to more locations) and the priority for new compute is to deploy more research assistants and scale research experiments.

Section 3. Compute Usage

BACK TO NAVIGATION

Status: Exploratory and uncertain, this section is a best guess not an informed forecast.

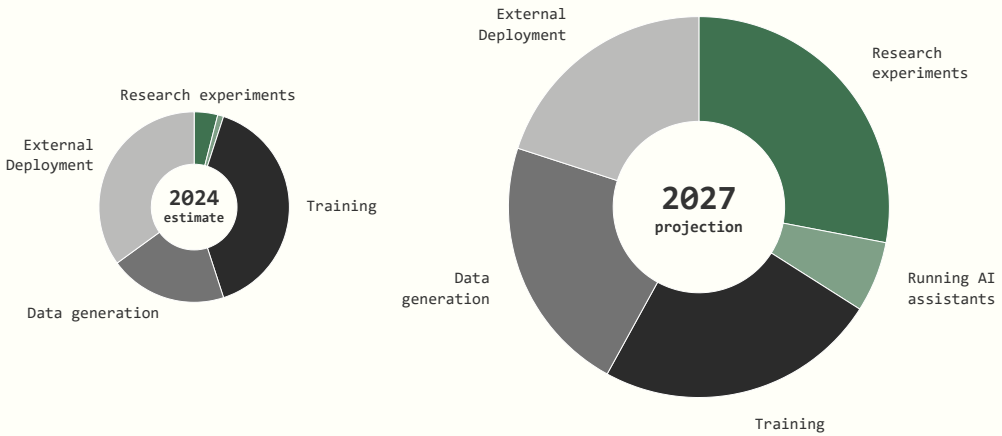


Figure 17: Summary of our compute usage forecast.

We expect the following internal compute usage breakdown for a leading AI company using public evidence from today's usage and educated guesses about their relative development, deployment, and research priorities going forwards. The rest of this section is

31. X dense-equivalent parameters means the inference cost is the same to a standard dense transformer with X parameters. For example, an 8-way MoE with 2T active parameters, 16T total parameters, would probably be about 8T dense-equivalent parameters.

32. Using the 6*parameters*tokens heuristic for training cost, and assuming 20% utilization in the post-training workload.

structured as a row by row justification for the values in the following table.

		2024	2025	2026	Q1 '27	Q2 '27	Q3 '27	Q4 '27 (racing)
Total compute usage and costs	Average compute used in during the period *(H100e)*	250K	1M	4M	7M	9M	12M	15M
	→ Compute spending during the period [26]	$5.4B	$12.5B	$30B	$20B	$25B	$30B	$45B
Breakdown	→ Compute budget in fp16 FLOP, 30% utilization	2.5e27	1e28	4e28	2e28	2.5e28	3e28	4e28
Training	**1) Training runs**	**40%**	**40%**	**40%**	**30%**	**27%**	**22%**	**20%**
	→ pre-training share	60%	50%	40%	10%	8%	5%	5%
	→ post-training	40%	50%	60%	90%	92%	95%	95%
Internal Workloads	2) Synthetic data generation	20%	20%	20%	27%	23%	22%	22%

		2024	2025	2026	Q1 '27	Q2 '27	Q3 '27	Q4 '27 (racing)
	3) Research experiments	4%	5%	6%	14%	21%	28%	35%
	4) Research automation	1%	3%	4%	6%	6%	6%	6%
	Total Internal	**25%**	**28%**	**30%**	**47%**	**51%**	**57%**	**63%**
	→ Capabilities share	95%	96%	97%	97%	97%	97%	97%
	→ Alignment share	5%	4%	3%	3%	3%	3%	3%[27]
External Inference	**5) External deployment**	**33%**	**30%**	**28%**	**20%**	**19%**	**18%**	**13%**
Other	**6) Monitoring**	**2%**	**2%**	**2%**	**3%**	**3%**	**3%**	**4%**

Total compute usage and costs

These rows are informed by our overall compute production and distribution projections to illustrate a typical leading AI company with a share of global compute usage growing from ~5% to ~20% from 2024 to 2027. In those sections we project the AI compute used by the leading AI company to grow 3.4x/year and computational price performance (FLOP/$) to improve roughly 1.4x/year. Read more on cost projections in the financials section.

33. Another reason why the external

Compute usage on training runs

In 2024 we estimate that OpenAI used an average of 40% of their compute on training. We expect most (around 60%) of this to be on trying to scale to next-generation pre-training runs, and the rest (around 40%) to be on reinforcement learning based post-training, with a focus on reasoning and agency. This is consistent with them doing a single pre-training run of about 3e26 fp16 FLOP of training, which corresponds to 100k H100s being used for 4 months, matching their historical scaling trend, and an average of 100k H100s being used on RL workloads for 2.5 months.[28] OpenAI seems to have relatively centralized access to a single cluster of about 100k H100s, and given that they are sourcing their compute from at least three providers in Coreweave, Oracle and Microsoft, it is unlikely that chips are centralized and well connected enough between different datacenters to enable a larger training run than this.[29] Furthermore, they have large compute requirements in internal and external deployment that more reasonably account for the rest of the total.

Going forwards we expect:

- Usage on training to fluctuate around 40% over the next two years.

deployment share might decrease is that as higher sensitivity capabilities are reached (such as scalable AI cyber defense agents) these may be marketed to corporate and institutional customers, which seem likely to be willing to pay high prices for early-access in special pre-release periods.

- Increasingly large shares of training compute are dedicated to post-training.
- In 2027, there are no large pre-training runs and training compute is directed to almost purely post-training workloads of the large model trained in 2026. [30]

Synthetic Data Generation

In 2024, we expect roughly 20% of OpenAI's compute to be used on generating synthetic data, the main use case probably being on eliciting the reported 'Orion' model as well as weaker base models with inference time techniques (e.g., tree search based) and using grading and filtering (e.g., rejection sampling) to then produce data that is used for post-training models like o1 and the recently announced o3. This is an average of 50k H100s used throughout the year or 2.5e26 fp16 FLOP. An example of how this could be used is for 5T forward passes on Orion (assuming it is a 2T parameter dense-equivalent model served at fp8 precision), [31] and 500T forward passes on GPT-4o. We'd guess the Orion forward passes are not filtered and are directly used as 5T tokens of high-quality training data, while the GPT-4o forward passes are roughly 16:1 in producing a high quality token (after search and grading) for another 30T tokens. This results in 35T total tokens generated, perhaps for post-training o1 and o3, which would allow for ~1e26

FLOP of GPT-4o post-training.[32] We expect synthetic data generation to become increasingly important in tandem with post-training workloads, staying at around 20% and then growing to ~30% in 2027.

Research experiments

We expect research experiments to have been a significant priority in 2024, with an average of 10k H100s being used on average throughout 2024. This corresponds to about 1e26 fp16 FLOP in total, which is sizable and will have allowed a range of small and large training and architectural experiments. As the emphasis shifts from training runs and external deployment to AI R&D automation, particularly in 2027, we expect this to grow steadily as the growing algorithmic research effort requires an increased share of the experiment budget. With research labour becoming increasingly automated, experimentation compute will become an increasingly important bottleneck to progress, which is why project usage will spike from 4% in 2024 to 15% in early 2027 and 35% in late 2027. **This would correspond to more than 2e28 fp16 FLOP of experiment compute in 2027.**

Research automation

34. This is the amount of data that can be moved onto SRAM per second (bidirectional).

35. Includes full server cost (e.g., networking, etc.) but excludes datacenter infrastructure (e.g., cooling and energy).

36. Average between SXM and NVL variants.

37. Based on online rumours of 12 HBM4E stacks per GPU.

38. Projected as a 2027 release that incorporates 4 more HBM4 stacks per GPU, possibly giving up some logic area in favor of more bandwidth.

We expect internal research automation to have been minimal in 2024, limited to employee early access use of models like o1 and o3 to help with research tasks. An average of 1k H100s, or 2e25 fp8 FLOP would be enough for 250B forward passes on Orion (using our best guess of it being 2T parameter dense-equivalent) or models like o1 or o3 which may have similar inference costs. As a sanity check, assuming 1000 employees, this amounts to 250M tokens per employee, or approximately 800k tokens per day. At 8:1 chain-of-thought ratios and 100 tokens/sec throughput, this represents about 1000 seconds or 16 minutes of generation time per employee per day, which seems like a reasonable average. Going forwards, as internal models become more capable, we expect them to scale this significantly, both in the number of copies of the model they run per employee, and in the scaffolding regimes (and therefore inference-time compute multiples) they typically deploy them at. This spikes particularly in 2027 with the push to automate their R&D workflows using expensive models from 1% today to around 6%, but doesn't go higher given the abundance of research labour and lack of experiment compute this would cause.

Compute usage on external deployment

We expect OpenAI to be using roughly 30% of their compute on external deployment through 2024 which is an average of 75k

39. Includes full server cost (e.g., networking, etc.) but excludes datacenter infrastructure (e.g., cooling and energy).

40. Has significantly less memory capacity, so while having extreme bandwidth it can only serve models up to ~40B parameters in size on a single chip. This can be solved by adding chip-to-chip interconnects to serve larger models with pipeline parallelism. Typical chip-to-chip bandwidth today enables efficient inference on Llama 405b across an estimated 10 or so chips.

41. ~~Projection~~ based on an optimistic 60% improvement in SRAM density due to TSMC's 2N node and possibly improved packaging techniques.

42. Since this is an in-house chip, there is no chip design margin, so this is similar to the direct manufacturing cost. We expect direct manufacturing cost to be around $400k, of which the vast majority is the cost of the HBM memory.

43. 360x12-Hi HBM4e stacks each with 2TB/s bandwidth and 24GB of memory, stacked directly on top of logic with 3D packaging.

H100s throughout the year. If they achieve an average of 10% inference utilization (across output and input tokens) on severed models (where it is expected to be difficult to batch requests) this corresponds to about 800T GPT-4o forward passes, or ~2T tokens per day.

This seems roughly right given that in February 2024, Sam Altman tweeted that OpenAI was generating 100 billion words per day. Assuming this roughly tripled by the middle of the year (slightly outpacing revenue with margins tightening slightly) to 400B tokens and roughly quadrupled with the release of o1-preview, o1, and o1-pro which use a disproportionate amount of inference compute to 1.6T tokens, assuming a 1:1 average input:output tokens ratio (given document and other inputs), this puts us at an average of roughly 1T tokens each per day throughout the year, for 2T total.

Going forwards we expect:

- The share should stay roughly the same through 2025 and then start to decrease as internal priorities grow and the release of frontier models in 2027 are delayed due the capabilities enabling significant productivity boosts on internal AI R&D efforts.
- Throughout the next three years, we expect revenue shares to shift increasingly towards corporate customers, as the business

44. X dense-equivalent parameters means the inference cost is the same to a standard dense transformer with X parameters. For example, an 8-way MoE with 2.5T active parameters, 20T total parameters, would be about 10T dense-equivalent parameters.

model shifts from being dominated by the 'online chatbot' to a 'drop-in remote worker service'. [33]

- Note that even a decreased share will still be a very large absolute jump, from 75k H100s in 2024 to 2M in 2027, which is roughly proportional to the increase in revenue we expect.

Compute usage on monitoring

As a very rough heuristic, we expect a model with an average inference cost around 10x lower than the average inference-cost of deployed models to be checking roughly half of all input/output tokens in various settings (either monitoring external deployment or, in later years, in potentially broader AI control setups, including for research automation). Therefore we add the external deployment and research assistant shares and divide by 20 to get our estimate for the share of compute on monitoring, and see that it stays roughly constant throughout at around 2-4%.

45. 2e28 / 2e25.

46. GPT-4o is estimated to have 200B active parameters. We scale these active parameters by sqrt(1000) to estimate ~6.3T active parameters for Agent-2. For simplicity, we use 10T in the calculations below and ignore whether this ends up being dense, or MoE (or even some future MoE-like architecture) as we expect

Section 4. AI research automati

BACK TO NAVIGATION

Status: Uncertain but informed. This section is an informed forecast based on public information.

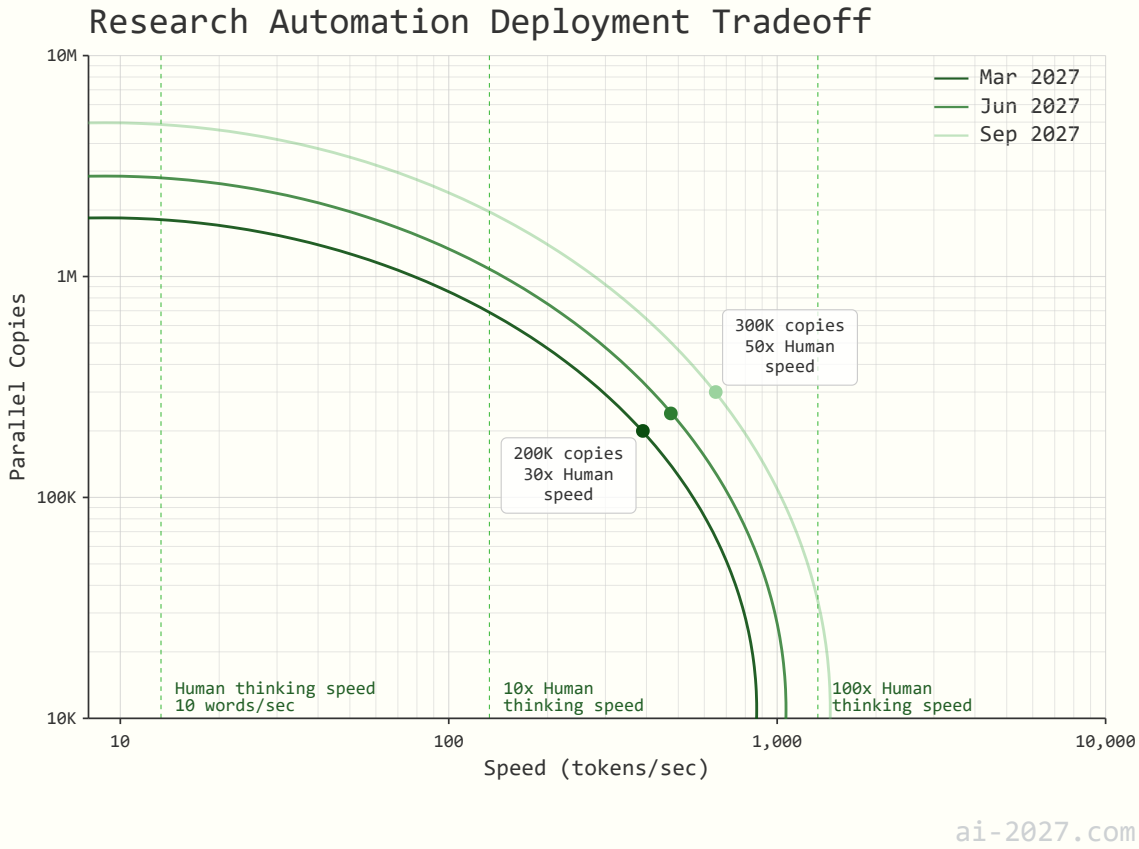


Figure 18: Summary of the AI research agent deployment tradeoff we expect OpenBrain to face using 6% of their H100e compute, as forecasted in the research automation compute usage section.

This subsection contains an analysis on the inference tradeoff between speed and parallel copies when serving an AI model, focusing on concrete AI models that we think might be used for AI research automation in AI 2027. In inference workloads, total

that a good routing setup could make this roughly equivalent to serving a 10T dense model.

47. We project an extreme degree of algorithmic efficiency progress especially spiking at the end of the year (in the racing ending). We expect this may approach the 'limits' of intelligence. Given the human brain is estimated to be a very sparse 100T fp8 parameter neural network, we expect it to be around 1T fp8 dense-equivalent parameters (implied by optimal training with 1e23 - 1e25 FLOP). We expect the human brain to be well within the limits of algorithmic efficiency in intelligence and learning.

aggregate memory bandwidth is the most direct measure of performance because decoding tokens sequentially requires maintaining a growing KV cache in fast memory and loading it, along with model weights, into processors for each token generated. Therefore, in this section we work in **H100-bandwidth-equivalents (H100-Be)**, so units of 3.6TB/s of total aggregate bandwidth (anchored to halfway between the H100 NVL and H100 SXM assuming these were produced roughly evenly).[34]

Existing Hardware Roadmaps

Nvidia GPUs

	GPU bandwidth	H100-Be	Memory capacity	Memory generation	Peak usage	Cost of ownership[35]
NVIDIA H100[36]	~3.6 TB/s	1	90 GB	HBM3	2023-2024	$40k
NVIDIA H200	4.8 TB/s	1.3	141 GB	HBM3e	2024-2025	$40k
NVIDIA B200	8 TB/s	2.2	192 GB	8x8Hi HBM3e	2025-2026	$50k
NVIDIA R200 Ultra[37]	19.2 TB/s	5.3	768 GB	12x16Hi HBM4	2027	$100k

48. This is currently typical as seen on OpenAI's API.

49. 11e12 Bytes / (20K (avg. seq length) * 50k (hidden dim) * 250 (layers) * 0.1 (grouped query attention) * 0.5 (fp4) * 2 (key+value)) = 440.

50. 'Parallel copies' = 'model instances' * 'batch size'

51. 720 TB/s / 16 TB/tok = 45 tok/sec.

	GPU bandwidth	H100-Be	Memory capacity	Memory generation	Peak usage	Cost of ownership [35]
NVIDIA R300 Ultra [38]	25.6TB/s	7.1	1TB	16x16-Hi HBM4	2027-2028	$120k

52. 720 TB/s * 10 chips / 5.3 TB/tok = 1,350 tok/sec.

Wafer Scale Inference chips

53. See raw calculations here.

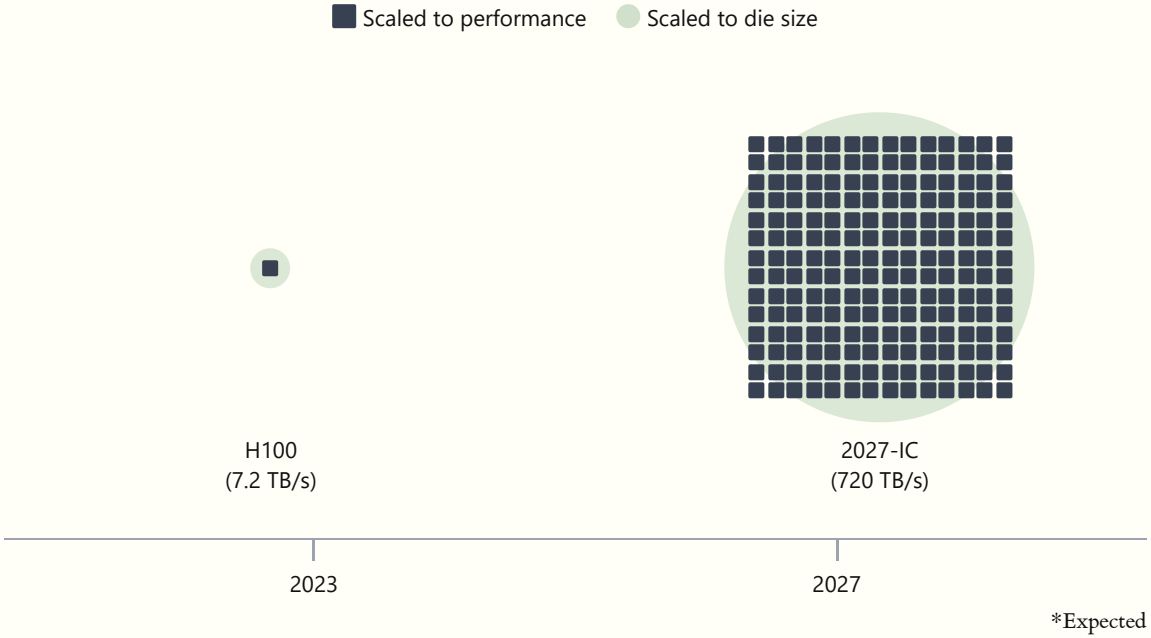
	GPU bandwidth	H100-Be	Memory capacity	Memory type	Peak usage	Cost of ownership [39]
Cerebras WSE-3	21,000 TB/s	5800	44 GB [40]	SRAM	2025-2026	$2-3M
WSE-2027 [41]	34,000 TB/s	9500	72 GB	SRAM	2027-2028	$3-4M

2027 In-House Inference Chip

This table contains a forecast on the **typical inference chip in 2027 (2027-IC)** that will be available to the leading AI company. As described in the hardware R&D automation subsection, we expect leading AI companies in 2027 to automate specialized chip designs and increase in-house chip production in 2027. There are early signs

of such ambitions, such as OpenAI's hiring of prolific Google TPU designers and plans to design their first in-house chip in 2025.

	TSMC process	Die size	H100e	GPU bandwidth	H100-Be	Memory capacity	Memory generation	Cost owners 42
2027 -IC	3nm	60xH100	100	720 TB/s	200	17.8 TB	HBM4e 43	$600k



		Q1 2027	Q2 2027	Q3 2027	Q4 2027 (racing)
Compute used on research automation	Research automation share	6%	6%	6%	6%
	Compute budget for research automation (H100e)	400K	600K	800K	1M
	Bandwidth budget for research automation (2027-ICs)	**4K**	**6K**	**8K**	**10K**
Breakdown	Bandwidth budget for research automation (H100-Be)	800K	1.2M	1.6M	2M
Deployment tradeoffs	Copies deployed (instances x average batch size)	**300K**	**400K**	**500K**	**600K**
	Avg. raw speed (forward	**230**	**290**	**360**	**430**

		Q1 2027	Q2 2027	Q3 2027	Q4 2027 (racing)
	passes / second)				
AI R&D progress[44]	Model size (parameters, fp8, dense-equivalent, average across deployed models)	10T	10T	5T	2T
Effect on AI R&D Progress (average during period)	Research Capability Level Milestone	Superhuman Coder	Superhuman Coder	Superhuman AI Researcher	Artificial Superintelligence
	AI R&D Progress Multiplier	**4x**	**10x**	**50x**	**2000x**

Frontier Model Size and Architecture

In early 2027, we expect frontier training runs to have reached around 2e28 FLOP (4e28 FLOP at fp8 precision) based on the compute usage estimates. Based on the ratio to GPT-4 ~1000x, [45] we naively predict that such an AI model would have roughly **10T active parameters.** [46] Given the degree of AI research automation

predicted by AI 2027, we expect significant distillation effects to reproduce the same level of capabilities in much smaller more efficient models. Our expectation is that AI companies will deploy a whole range of both smaller models, and new checkpoints of their largest model which they are continuously post-training to higher capabilities. Nonetheless, we perform our speed and copies calculations in units of a **fixed 10T parameter dense-equivalent model** for simplicity, and assume that mixture of experts or other future architectures can be converted into 'dense-equivalent' parameters based on their inference economics. We then adjust speed/copies calculations in future periods according to the decreasing model size. Notably, we expect the most intelligent model to fall from around 10T dense-equivalent parameters to 2T parameters by the end of 2027, even as the qualitative ability of the model increases from a median human researcher to a vastly superintelligent researcher, due to algorithmic efficiency exploding and nearing the limits of intelligence. [47]

Compute and memory available

According to our compute distribution projection the leading AI company in 2027 will have 15-20% of the world's compute and according to our internal usage projection they will use 6% of this on running AIs for research automation, giving them around 4K

2027 In-House Inference Chips (2027-ICs) in bandwidth budget for Q1 2027, and 10K 2027-ICs by Q4 2027. We perform calculations in terms of **1K 2027 In-House Inference Chips** (1k 2027-ICs) equivalent to 100K H100s in compute and 200K H100s in bandwidth and then use to back out Q1 to Q4 deployment tradeoff curves.

Trading off parallel copies and speed

When running inference you can trade off running more parallel copies with slower speed and vice versa, where the trade-off is limited by memory bottlenecks (when scaling up copies) and latency bottlenecks (when scaling down copies, maximizing speed). In other words, if you scale up parallel copies beyond what fits in the HBM total memory of your GPU server, or group of well connected GPU servers, you would start to see significant slowdowns from having to use slower memory bandwidth connections, while if you scale down too aggressively, by reducing the batch size, and/or by increasing tensor/model parallelism (which means spreading the model across more GPUs), each request needs to travel across more chips in a cluster, leading to communication bottlenecks.

Scaling up parallel copies

54. If a larger company like Google or Meta become the leading AI company, we are projecting their AI-only revenue and costs in this section, treating their AI efforts as an imagined sub-entity similar to the AI-only startups like OpenAI, Anthropic and xAI.

Assuming inference is run at fp4 precision, 10T parameters would require 5TB of memory. On one of the 2027-IC chips, this would leave 12.8TB of HBM free, of which we might assume 1.8TB is reserved for miscellaneous purposes, and 11TB is left for the KV cache. Assuming an average sequence length of 20k tokens, [48] a hidden dimension of 50k, and 250 total layers, this could support a maximum batch size of 440. [49] So with 1000 instances (1 per 2027-IC chip) this would correspond to **440K parallel copies.** [50] Each batch would require 16TB/tok of bandwidth, so the inference speed would be **45 tokens/second.** [51]

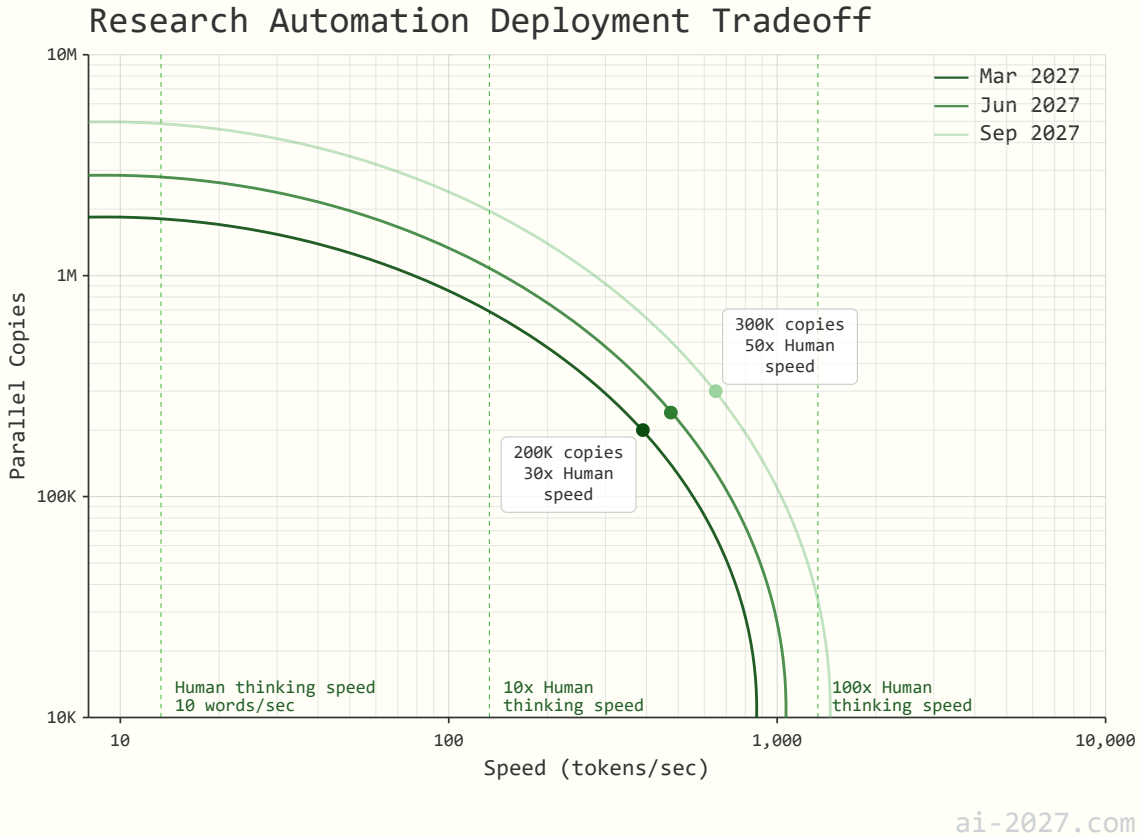
Scaling down parallel copies

In theory you could scale down to just 1 model instance across all 1000 servers, but this would suffer majorly from communication bottlenecks. More reasonably, we might assume that you would have a model instance spread over 10 2027-IC chips and have a batch size of just 10. This would leave you with just **1,000 parallel copies**. Each batch would require 5.3 TB/tok of bandwidth, so the inference speed would be **1,350 tokens/second,** [52] assuming sufficient interconnect between the 10 chips for there to be no inter-chip communication bottleneck.

Projected Deployment Tradeoff

55. The only other significant datapoint is the B200, which is supposed to cost 1.3x more than the H100 with 2.5x the computational performance. For a net 1.9x increase in FLOP/$ over ~20 months. This is a yearly rate of 1.5x/year. Naively looking at the 2nd derivative between the

Putting together these two 'scale up parallel copies' and 'scale down parallel copies' extremes, we can compute a range of deployment choices in between to get average deployment tradeoff curves for each time period as shown in Figure 16.[53] These curves were not computed with full consideration of possible current or future parallelism techniques and chip-to-chip communication bottlenecks, though we are relatively confident that these would be unlikely to affect the curves significantly.



A100 and the B200, we can guess that the next doubling in FLOP/$ will take 24 months, which would average out to a trend of around 1.5x/year. Under the capabilities projections in AI 2027 we expect competition over AI chips to increase dramatically in 2027, leading to higher pricing and therefore only a 1.2x FLOP/$ improvement in 2027 (with high uncertainty). This all averages to around 1.4x/year for the next three years.

Figure 19: AI research agent deployment tradeoff we expect OpenBrain to face using 6% of their H100e compute, as forecasted in the research automation compute usage section.

We naively speculate that through 2027 there will be a roughly even preference for high parallel copies and high speed with the deployment tradeoff averaging to around 300K average copies and an average speed around 20x human thinking speed. Out of the scope of this piece, we expect the preference to shift slightly towards serial workloads in 2028 as there are increasingly diminishing returns to more copies of the model towards the limits of intelligence, and new hardware inventions from mid-late 2027 have time to be manufactured and widely in-use.

	Q1 2027	Q2 2027	Q3 2027	Q4 2027
Parallel copies	300K	400K	500K	600K
Speed	230 tok/sec	290 tok/sec	360 tok/sec	430 tok/s
parameters	*10T*	*10T*	*5T*	*2T*

56. FP16 petaFLOP/s / kW

57. Based on applying a 25% efficiency gain over the B200.

Section 5. Industry Metrics

BACK TO NAVIGATION

Status: Uncertain but informed. This section is an informed forecast based on public information.

This section presents a projection of leading AI company frontier training runs through 2028, and discusses the associated projections in costs, revenues, and power requirements.

Training runs

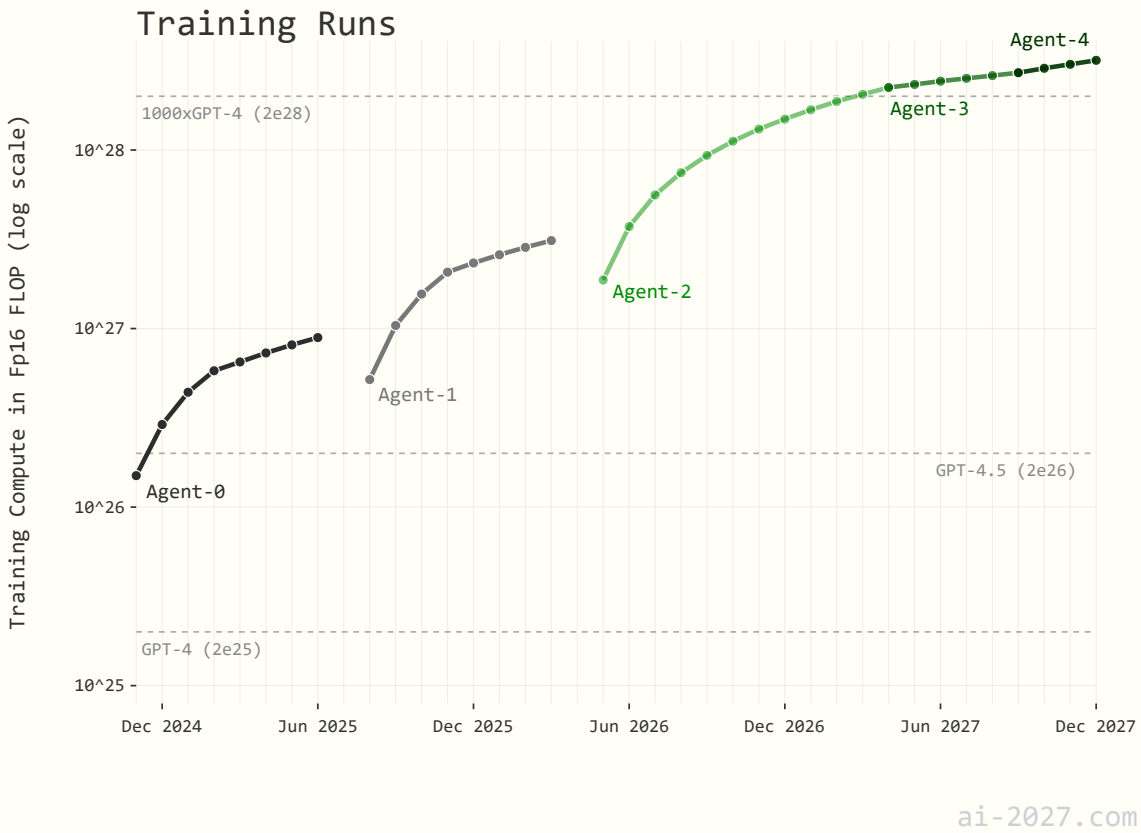


Figure 20: The training runs in AI 2027.

Our compute production forecast has the global AI-relevant compute growing at 2.25x per year between 2024 and 2027. With the leading AI company's share of the global stock growing 1.4x per year, and their internal usage on training runs staying at 40% until 2027 when it drops to 20%. Putting this all together we project the following training runs.

Model	Training period	Global compute during training period (H100e)	Share of global compute	Share of internal usage	Training compute
Agent-0	Oct 2024 - May 2025	10M	6%	40%	1e27
Agent-1	Jul 2025 - Feb 2026	18M	9%	40%	4e27
Agent-2	Apr 2026 - Mar 2027	38M	14%	36%	2e28
Agent-3	Mar 2027 - Aug 2027	60M	16%	24%	+1e28^
Agent-4	Aug 2027 - Dec 2027	80M	18%	20%	+1e28^^

Chips in use

Using our compute production and compute distribution sections, we illustrate what we expect to be the chips to be in use by the leading AI company through 2027. We expect the bulk to be Nvidia chips along with in-house inference chips emerging in 2027.

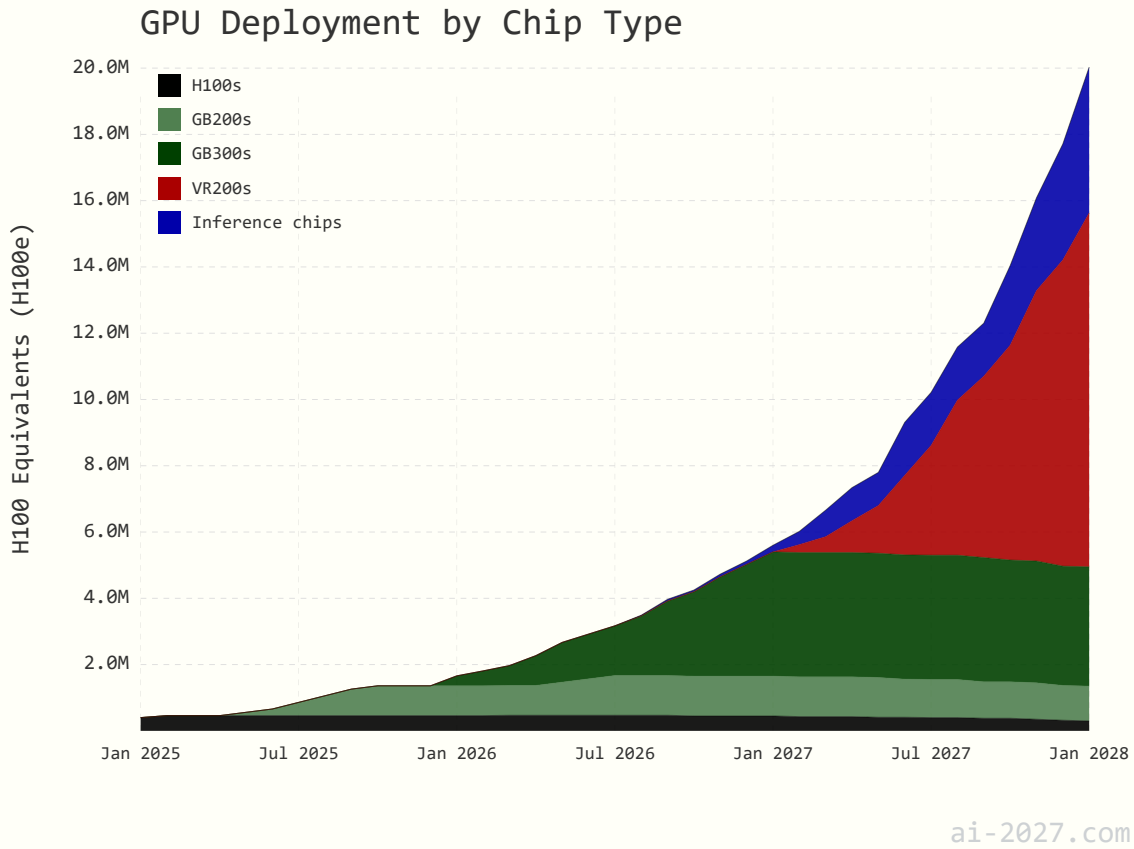


Figure 21: Chips in use by the leading AI company through December 2027.

Financials

In this subsection we project the leading AI company's costs and revenues through 2027.[54] We anchor on OpenAI to understand the pre-2025 trend and then use our other projections to extrapolate forwards.

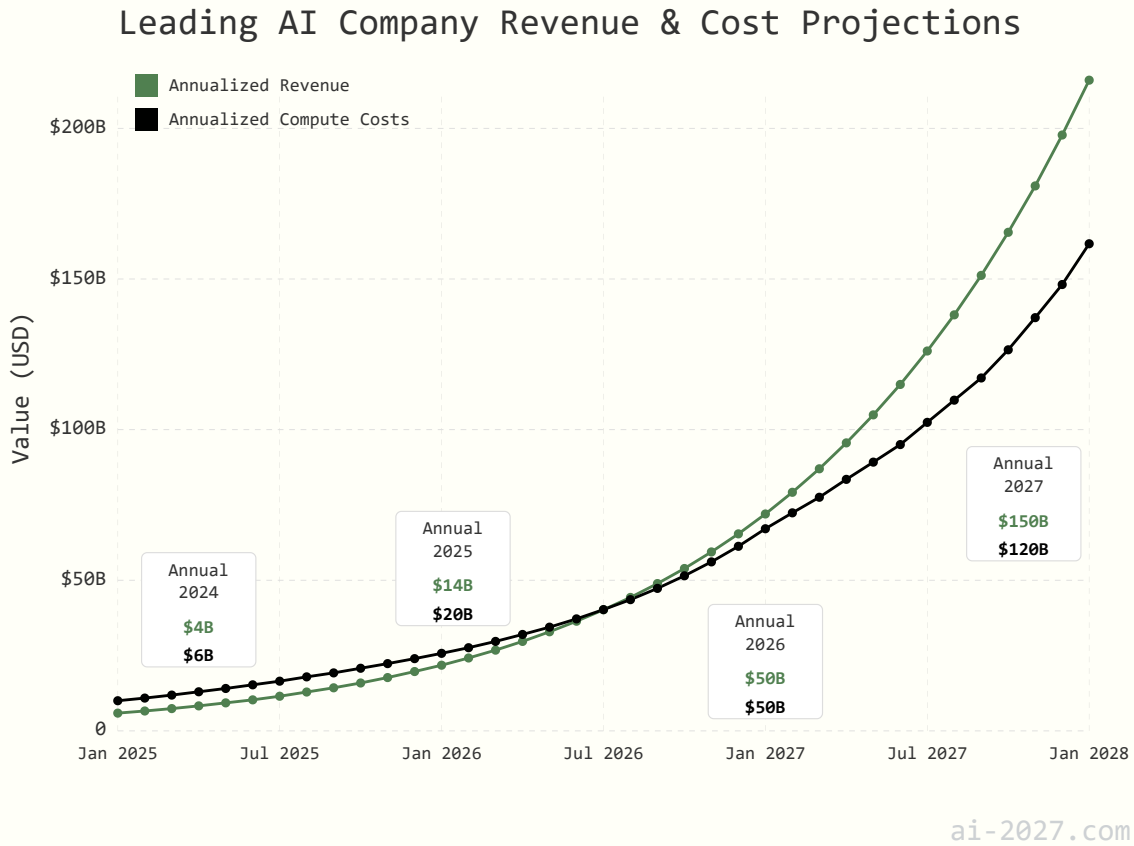


Figure 22: Approximate cost and revenue projections for OpenBrain, the leading AI company.

Compute cost projection

The New York Times reported that OpenAI anticipates $5.4B in computing costs in 2024, and GPT-4 which was trained in mid 2022 over 3 months, is estimated to have cost around $100M to train. Assuming GPT-4 training was 20-40% of their compute spending that year, that means they saw a roughly 11-22x/year increase in their compute spending from 2022 to 2024.

Meanwhile, computational price performance (FLOP/$) doubled from the A100 to H100 in 16 months, which we expect to have been a slight outlier to a trend closer to around 1.4x/year on frontier AI chips.[55] Going forwards, we can take our 3.4x/year projection for a leading AI company's compute and adjust down by 1.5x improvement in FLOP/$ to get an average **2.4x/year projection in spending on AI compute by the Leading AI Company**. Though we expect this growth to be faster in the first year or two and then slow due to increasing competitiveness of in-house chip designs.

Revenue projection

We use OpenAI's 2023 revenue of $1B and 2024 revenue around $4B to to piece together a short term trend that we expect to slow down gradually, but see sustained exponential growth through 2027 due to agentic models attracting high paying subscribers for products such as 'drop in remote workers.'

	2023	2024	2025	2026	2027
Annual Revenue	$1B	$4B	$14B	$45B	$140B
Revenue Y/Y Growth	-	300%	250%	221%	211%
Annual Compute Cost	$1.8B	$6B	$16B	$40B	$100B
Compute Cost Y/Y Growth	-	233%	166%	150%	150%

FutureSearch projected how OpenBrain's revenue might grow to reach $100B in mid-2027. They looked at the overall speed of the fastest companies to scale from $1B to $100B and found that the time to make that transition has been decreasing over time. A leading company making the jump in only 4 years would be the fastest ever, but also is on trend. They also examined the types of Agents that might be in highest demand - from consumer service representatives to R&D researchers - along with their price points, to produce a detailed sketch of where OpenBrain's revenue most likely would come from. For more see FutureSearch's full report.

Power requirements

Finally, we model the power requirements for the compute used by the leading AI company through 2028 based on reported and projected power efficiency of the key Nvidia chips we expect to be in use. Using this we can back out the total power required by the leading AI company as well as the implied total AI datacenter power used globally (assuming the leading AI company has an average power usage efficiency ratio).

	Performance (fp16, FLOP/s)	Peak usage	Datacenter peak power usage	Energy efficiency [56]
NVIDIA H100/200 GPU	1 x 1015	2023–2024	1000W	1.0
NVIDIA B100/200 GPU	2.5 x 1015	2025–2026	1700W	1.47
NVIDIA R100/200 GPU	6 x 1015	2027–2028	~3300W [57]	1.81

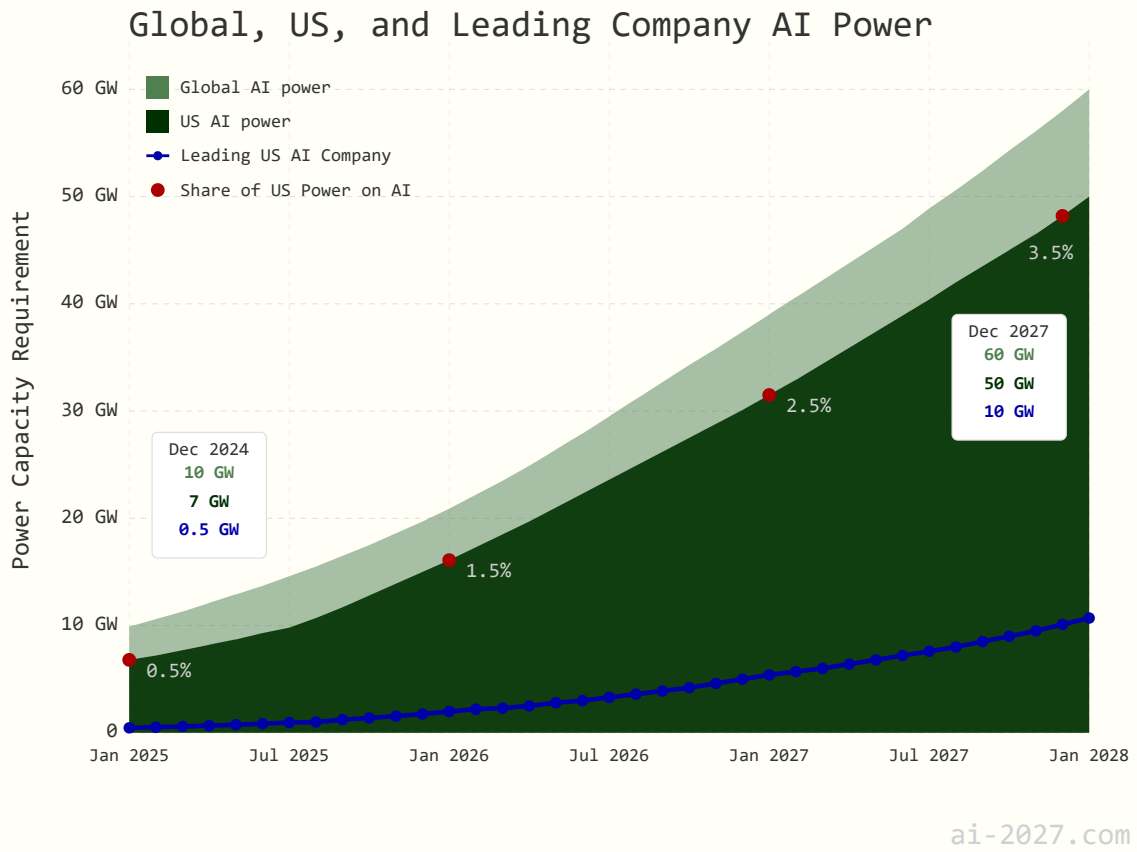


Figure 23: Power requirement projections through December 2027.